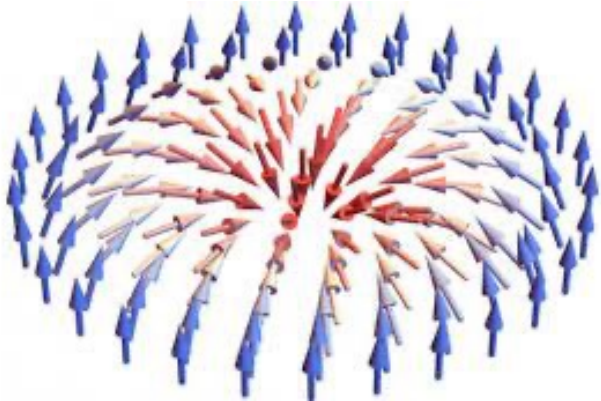


HOPFIONS IN CONDENSED MATTER & FIELD THEORY

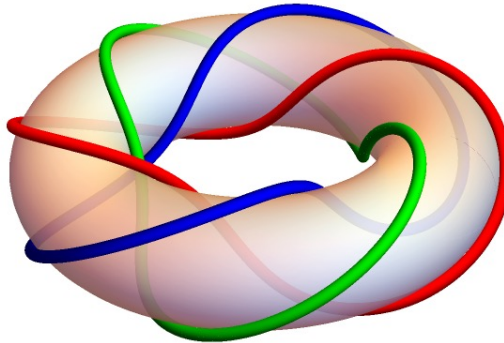
Radha Balakrishnan (The Institute of Math. Sciences, India), **Rossen Dandoloff** (Sofia University)

Avadh Saxena (Los Alamos National Lab)

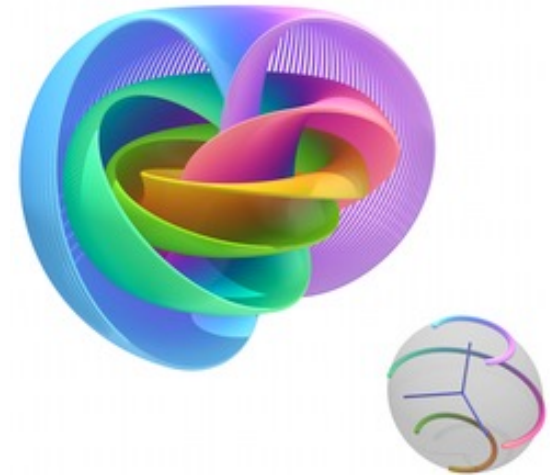
Phys. Lett. A 480 (2023), Phys. Lett. A 493 (2024)



1961 → 2009 (**Munich**, chiral magnet)



LA-UR-22-20781



1975 → 2017 (**Boulder**, liquid crystal)

Los Alamos, New Mexico, USA: altitude 7,300'





THEORETICAL DIVISION

MAY 10, 1945

H.A. Bethe, Division Leader, E-208, Ext 71
V.F. Weisskopf, Deputy Division Leader, E-217, Ext 74
J. von Neumann, Consultant, E-205, Ext 72 R3

GROUP T-1

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R.F. Christy, Section Leader E-120, Ext 178 R2
K. Fuchs, Section Leader, E-118, Ext 77
Baroody, E.M., E-121, Ext 178 R2
Calkin, J.W., E-117, Ext 77
Inglis, D.R., T-35, Ext 54
Keller, J., E-120, Ext 178 R2
Penny, W.G., E-101A, Ext 470
Podgor, T/5 S., E-116, Ext 469
Roberts, T/5 A.E., E-117, Ext 77
Skyrme, T.H.R., E-118, Ext 77
Stark, R.H., E-116, Ext 469
Stein, T/5 P.R., E-121, Ext 178

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L.I. Schiff, Alt. Leader, E-108, Ext 76
Case, K.M., E-107, Ext 76
Glauber, R., E-107, Ext 76
Kurath, T/3 D., E-202, Ext 205
Rarita, W., E-111, Ext 177
Richman, C., E-110, Ext 177
Stehle, T/5 P., E-107, Ext 76

GROUP T-3

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R.E. Marshak, Alt. Leader, E-218, Ext 74 R2
Bellman, Pvt. R., E-222, Ext 468
Cohen, T/5 S., E-219, Ext 74 R2
Lennox, T/5 E., E-218, Ext 74 R2
Olum, Paul, E-216, Ext 74
Smith, J.H., E-219, Ext 74
Wing, Milton, E-222, Ext 468
Bowers, W.A., E-216, Ext 74

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J. Ashkin, E-209, Ext 72 R2
Ehrlich, R., E-210, Ext. 72 R2
Peshkin, T/4 N., E-203, Ext. 79 R3
Reines, F., E-210, Ext 72 R2
Welton, T.A., E-209, Ext 72 R2

GROUP T-5

D. Flanders, Leader, E-205, Ext 69 R1
P. Whitman, Alt. Leader, E-204, Ext 69 R2
Atkins, A.L., E-211, Ext 73 R2
Davis, R.R., E-215, Ext 74
de le Vin, E., E-201, Ext 79
Elliott, J., E-213, Ext 73 R1
Hauser, T/5 F.H., E-214,
Huber, T/4 .C., E-120, Ext 70
Hudson, H., E-214, Ext 73 R1
Inglis, B., E-212, Ext 73 R2
Johnson, M., E-212, Ext 73 R2
Kellogg, T/5 H., E-203, Ext 69 R3
Langer, B., E-201, Ext 79
Page, T/3 W., E-214, Ext 73 R1
Rau, E., T/3, E-120, Ext 70
Staley, T/3 J., E-212, Ext 73 R2
Teller, M., E-214, Ext 73 R1
Vuletic, T/5 V., E-211. Ext 73 R2
Wilson, F., E-212, Ext 73 R2
Wright, T/5 E., E-213, Ext 73 R1
Young, T/Sgt., G., E-211, Ext 73 R2

GROUP T-6

E. Nelson, Leader, E-116
N. Metropolis, Alt. Leader, E-115, Ext 78
R. Feynman, Consultant, E-206, Ext 72
Ewing, F.E., E-112, Ext 75
Goldberg, T/5., E-108, Ext 76
Kemeny, Pvt. J., E-105, Ext 75
Hamming, R.W., E-114, Ext 78
Heermans, Corp. A., E-105, Ext 75
Heller, T/5 A., E-105, Ext 75
Hurwitz, T/5 D., E-105, Ext 75
Johnston, T/3 J., E-103, Ext 75
Kington, T/4 J., E-105, Ext 75
Livesay, N., E-112, Ext 75
Ninger, H., E-105, Ext 75
Noah, F.E., E-105, Ext 75
Vorwald, T/5 A., E-105, Ext 75
Zimmerman, T/3 W., E-105, Ext 75

GROUP T-7

J. Hirschfelder, Leader, T-30, Ext 206
J. Magee, Alt. Leader, T-30, Ext 206
Brummer, T/5 E., T-28, Ext 206
Feckete, T/4 P., T-26, Ext 206
Larson, T/4 L., T-28, Ext 206
Ostrow, E., T-30, Ext 206
Schwartz, T/4 P., T-28, Ext 206

GROUP T-8

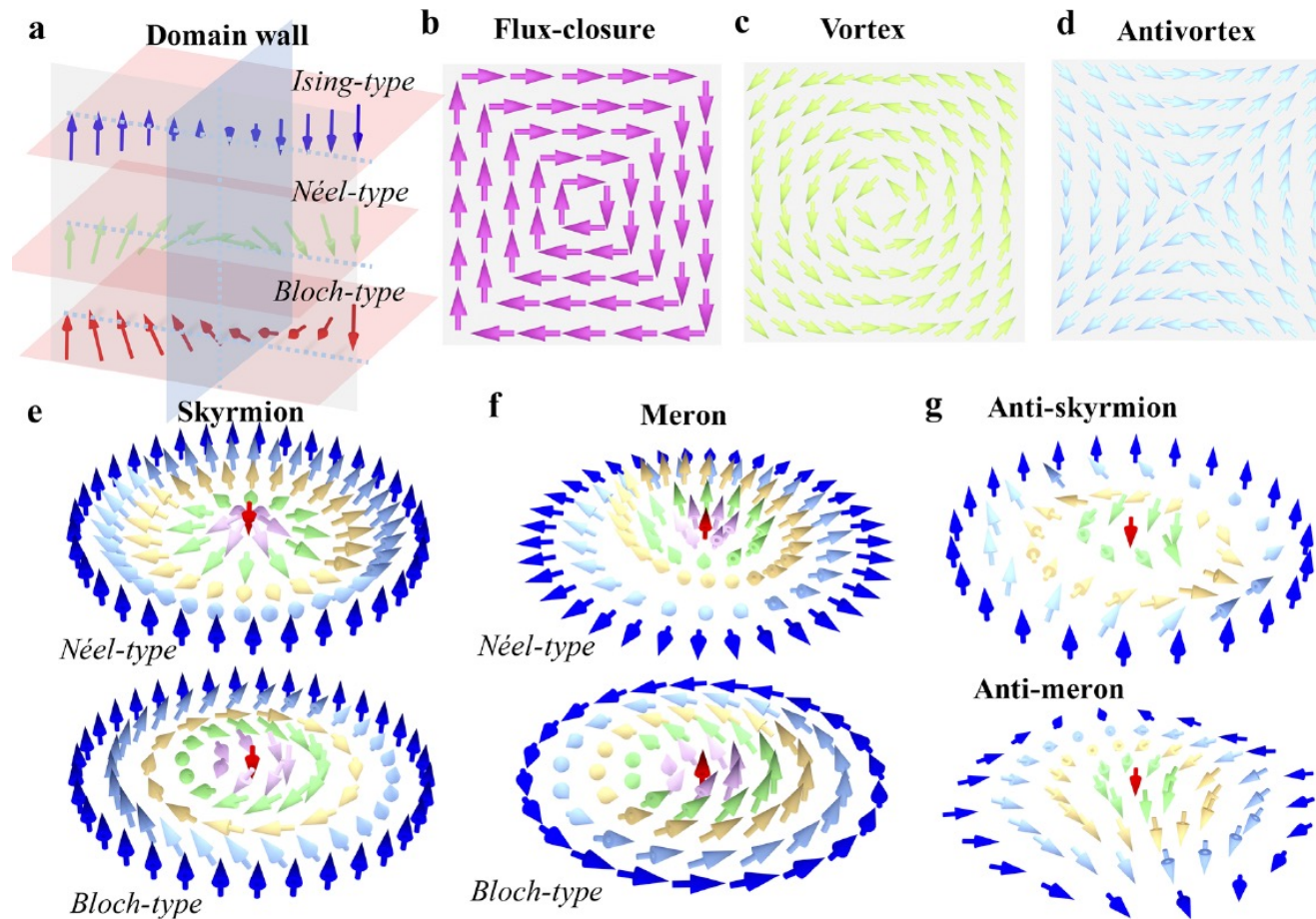
G. Placzek, Leader, E-220, Ext 468
Mark, C., E-221, Ext 468 R2
Carlson, B., E-221, Ext 468 R2
Day



SKYRMIONS & HOPFIONS in Different Vector Fields

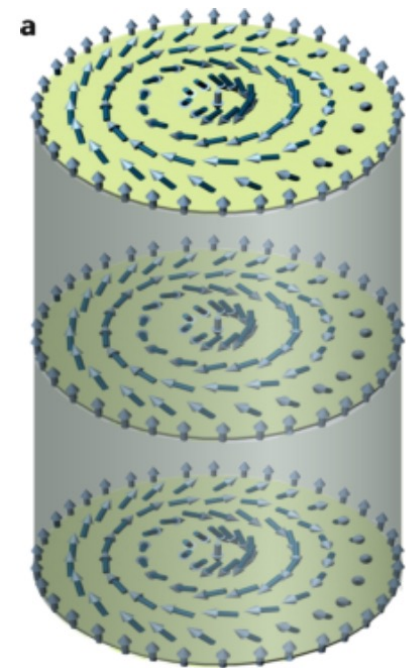
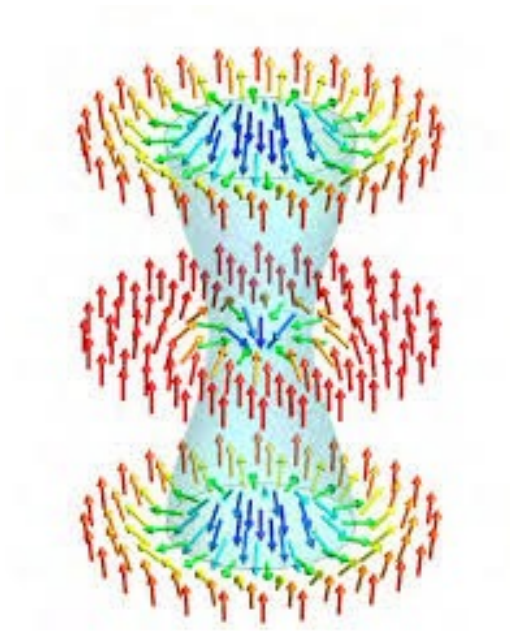
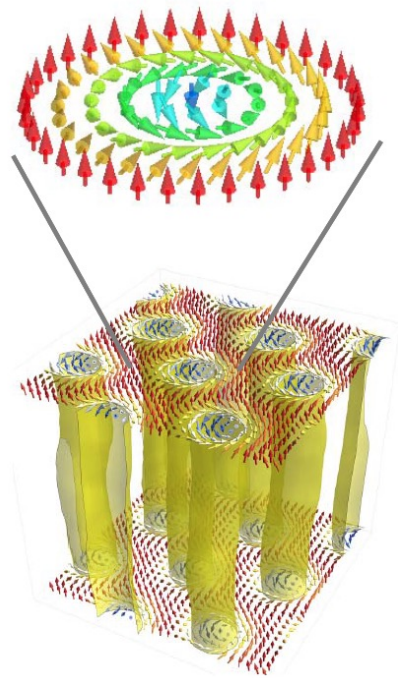
- **Magnets**: Axial vectors (Skyrmion and **Hopfion solutions**).
- **Ferroelectrics**: Polar vectors (with boundary conditions).
- **Liquid Crystals**: Directors (rods, no arrows).
- Bose-Einstein **Condensates**: Local spin bending angle.
- **Focus: Exact solution for Hopfions** in **anisotropic magnets**.
- **Chern insulators** and **Hopf insulators**; **Photonic hopfions**.

Topological Textures in Ferroic Materials



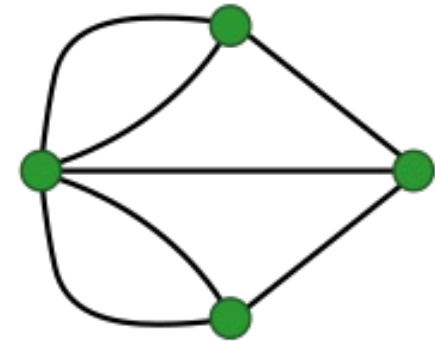
S. Das et al., APL Mater. **8**, 120902 (2020)

Skyrmion Tubes:

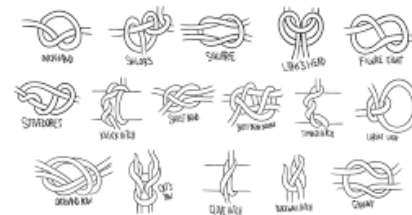
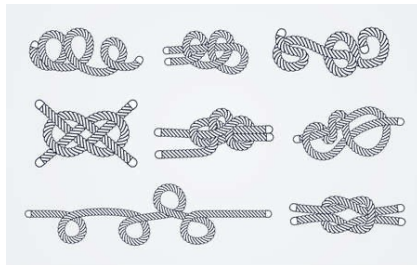


Historical perspective: Knots and topology

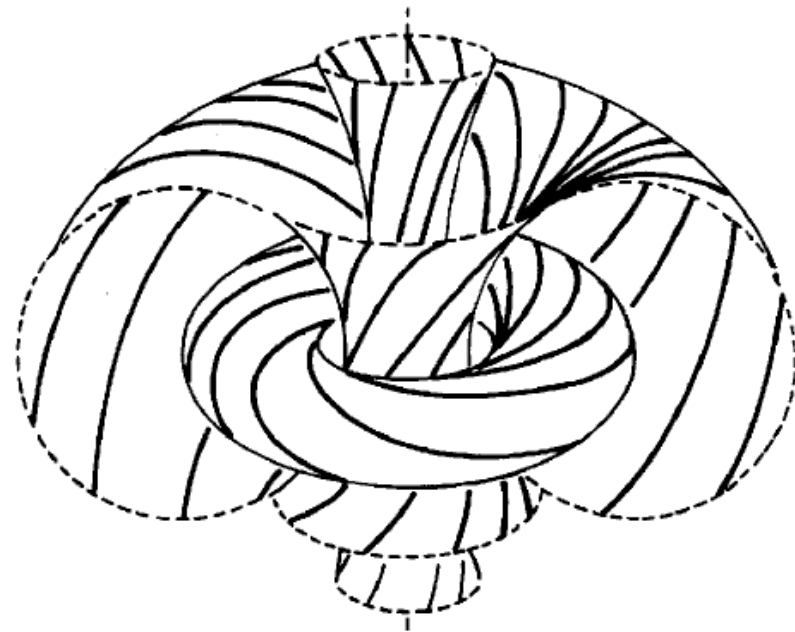
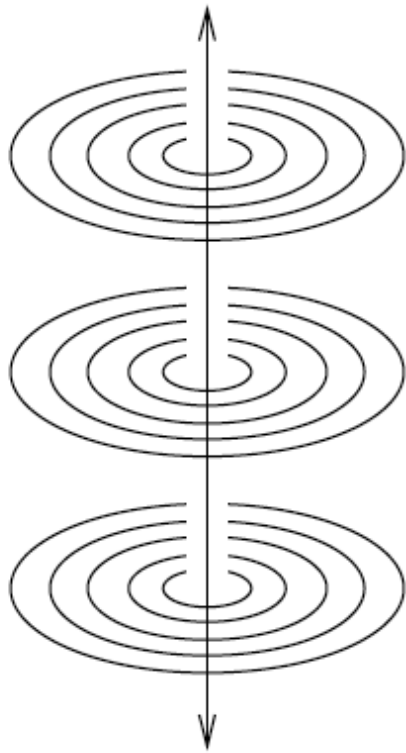
Leonhard Euler (1736): 7 bridges of Kaliningrad (led to graph theory and topology)



Lord Kelvin: In 1867 proposed atoms as knotted vortex tubes (in ether)



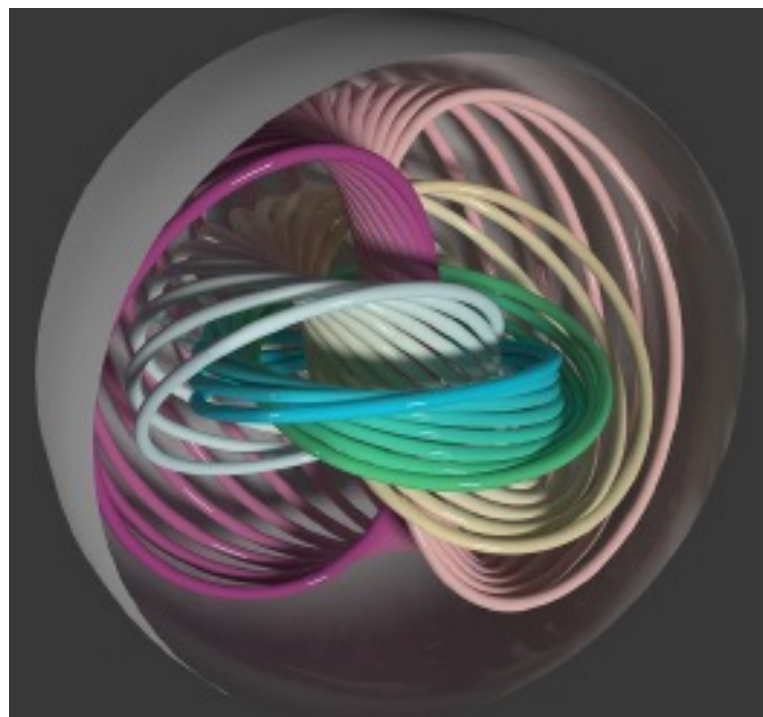
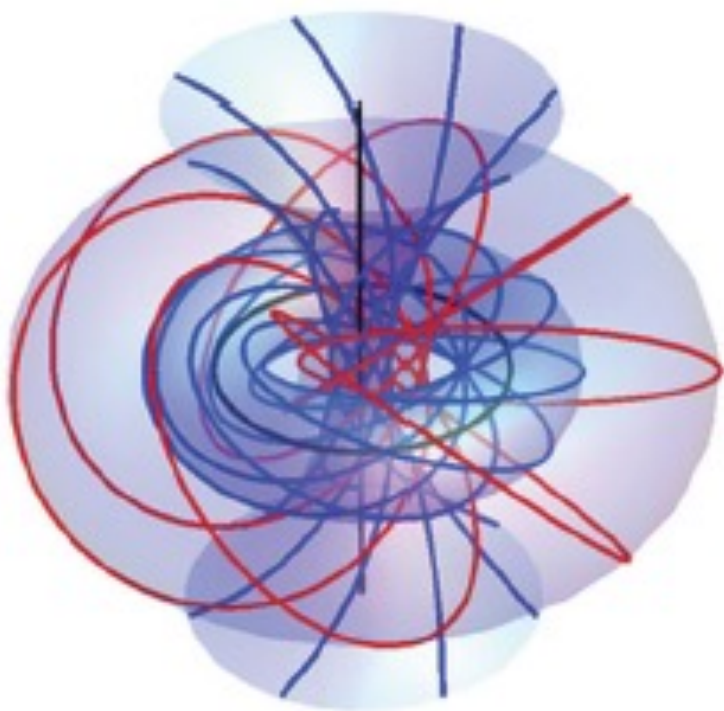
Trivial vs Nontrivial Fibrations of $\mathbb{R}^3$



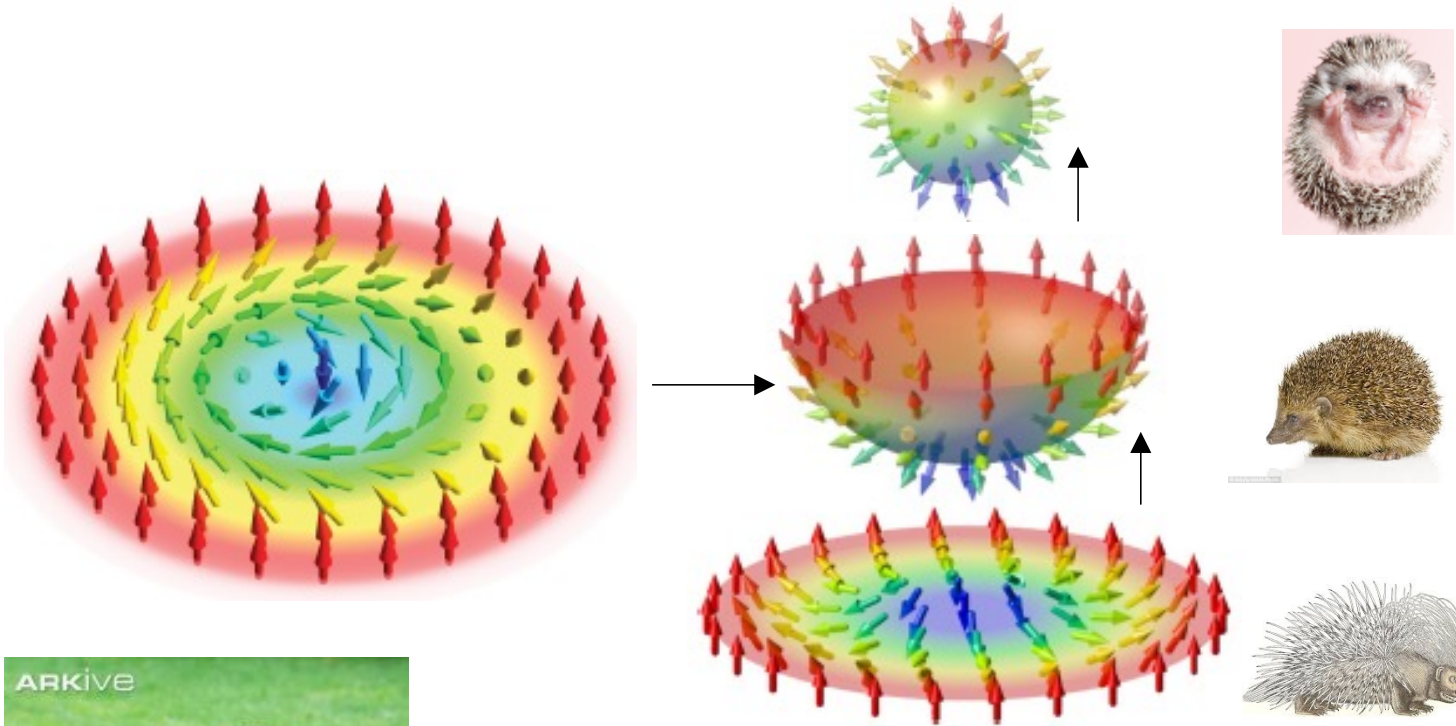
TRIVIAL (**unlinked** circles)

NONTRIVIAL (Clifford parallels: **Hopf** fibration)

HOPFIONS AS NESTED TORI



A SKYRMION “COVERS” THE SPHERE ONCE

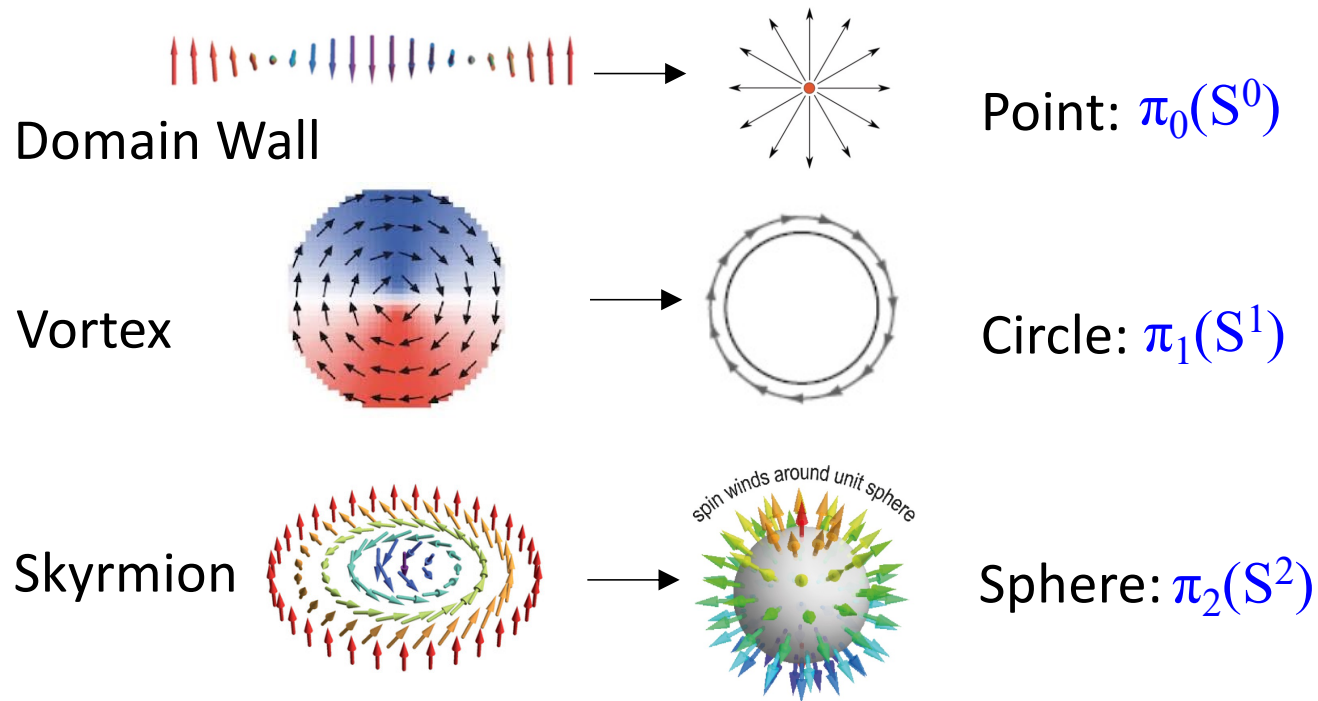


Tony H. R. Skyrme
1922 - 1987

Topological charge = 1 Hedgehog

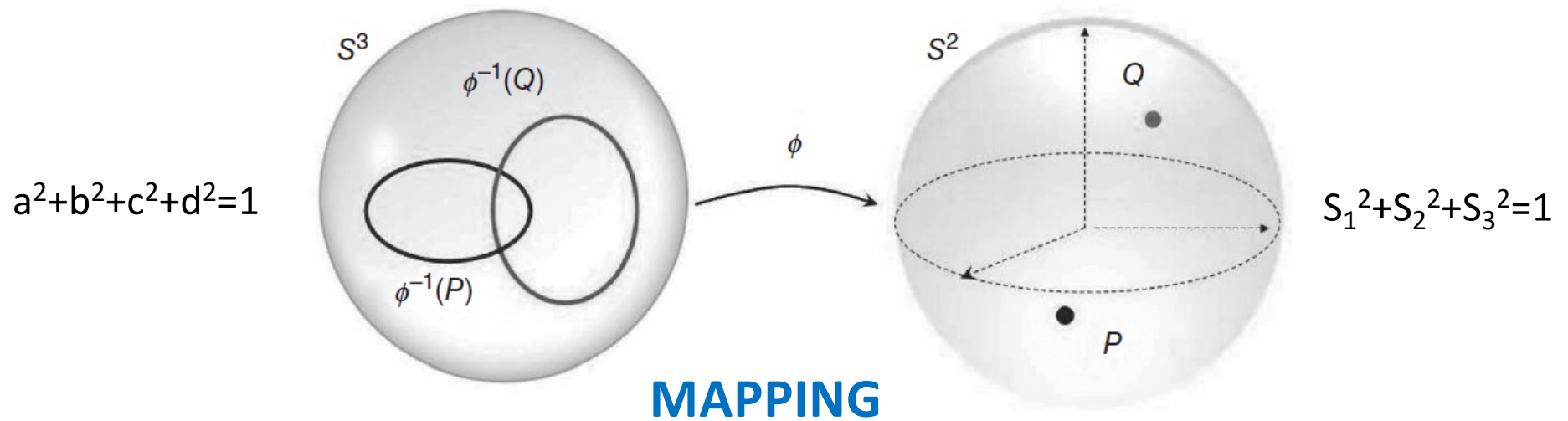
“Topologically Protected” Animal!

<u>Topological Excitation</u>	<u>Manifold Surface</u>	<u>Homotopy Class</u>
-------------------------------	-------------------------	-----------------------



Integer number of windings **Mapping:** real space to spin space

Hopf map: Preimages of two points P & Q



Hopfion:	$\Pi_3(S^2) : S^3 \rightarrow S^2$	(Linking number/Hopf invariant)
Skyrmion:	$\Pi_2(S^2) : S^2 \rightarrow S^2$	(Winding number)

$$S_1 = (a^2 + b^2) - (c^2 + d^2)$$

$$S_2 = 2(ad + bc)$$

$$S_3 = 2(bd - ac)$$

Hopfions introduced by Faddeev in 1975

- **Field theory model**: Skyrme-Faddeev or Faddeev-Niemi model
- Nonlinear sigma model plus an additional gradient term and a potential
- Various homotopy groups
- Pontryagin index or winding number:

$$\pi_1(S^1) = Z, \quad \pi_2(S^2) = Z, \quad \pi_3(S^3) = Z$$

$$\pi_1(S^2) = 0, \quad \pi_2(S^1) = 0, \quad \pi_3(S^2) = Z \text{ (Hopfion)}$$

L.D. Faddeev, "Quantization of Solitons", Preprint IAS Print-75-QS70 (Princeton, NJ)

L.D. Faddeev, "Some Comments on the Many Dimensional Solitons", Lett. Math. Phys. **1**, 289 (1976)

Faddeev-Niemi Model

$$H = E_2 + E_4 = \int d^3x [g^2 (\partial_\mu \mathbf{n})^2 + e^2 F^2]$$

Two-form: $F = (d\mathbf{n} \wedge d\mathbf{n}, \mathbf{n})$ on S^2

Pre-image: $F_* = dA_*$ on S^3

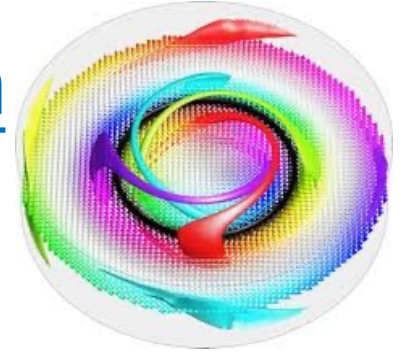
Hopf invariant: $Q_H = \int_{R^3} F \wedge A$

$$\mathbf{n} \cdot \mathbf{n} = 1$$



Ludvig D. Faddeev
1934 - 2017

Faddeev-Skyrme Model Lagrangian



3D vector field with $n^2 = 1$ leads to Hopfions

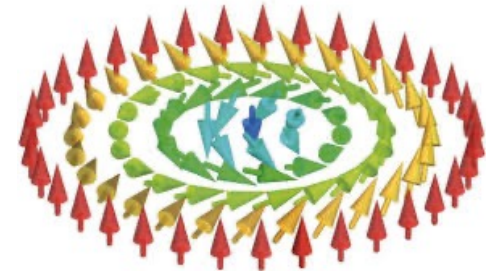
$$\mathcal{L}_{FS} = \partial_\mu \mathbf{n} \cdot \partial^\mu \mathbf{n} - \frac{1}{4} (\partial_\mu \mathbf{n} \cdot \partial^\mu \mathbf{n})^2 + \frac{1}{4} (\partial_\mu \mathbf{n} \cdot \partial_\nu \mathbf{n}) (\partial^\mu \mathbf{n} \cdot \partial^\nu \mathbf{n}) - V(\mathbf{n})$$

Map: $S^3 \rightarrow S^2$ (linking number)

2D vector field with a mass term leads to baby Skyrmions

$$\mathcal{L}_{SM} = \mathcal{L}_{FS} + M^2(n_z - 1)$$

Map: $S^2 \rightarrow S^2$ (winding number)

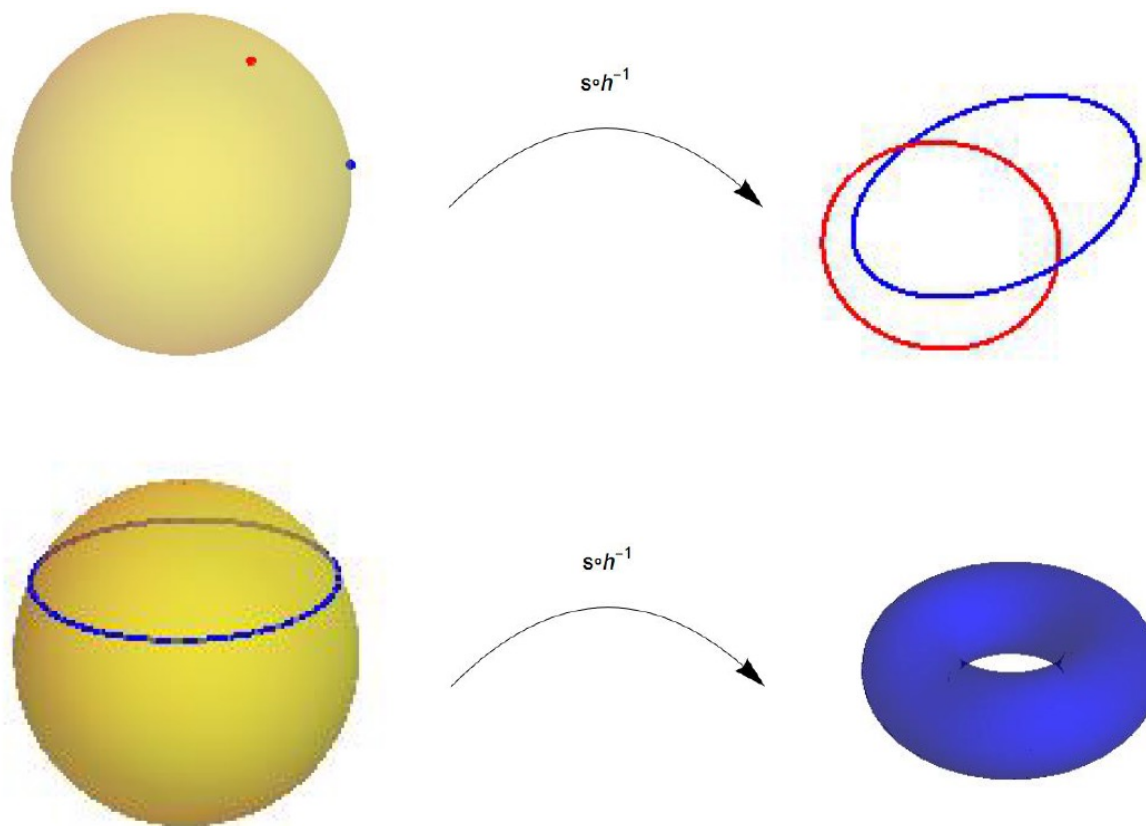


A Part of the Hopf Fibration

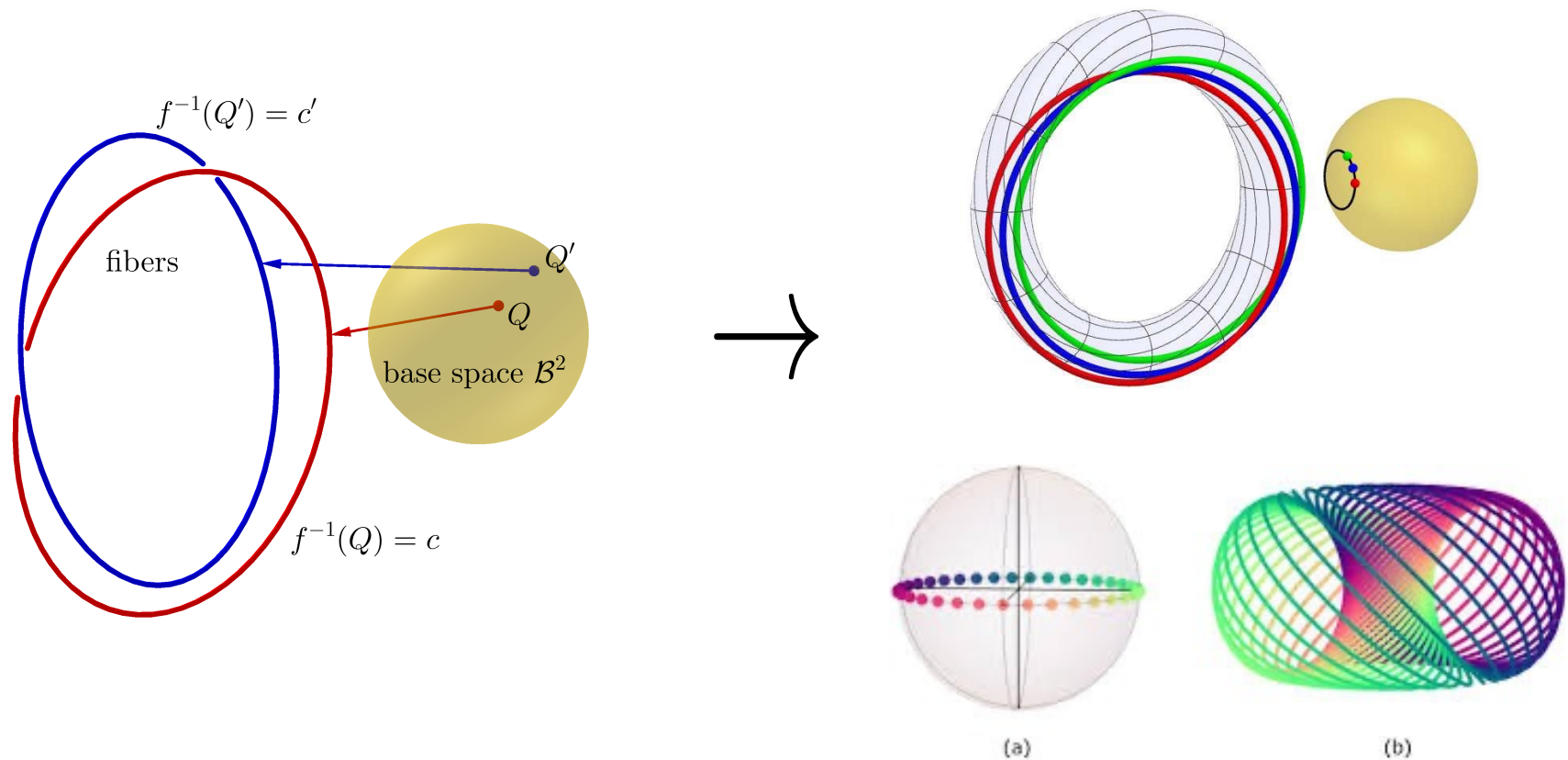


S^3 to S^2 Mapping is the Generator of the Third Homotopy Group of the 2-Sphere

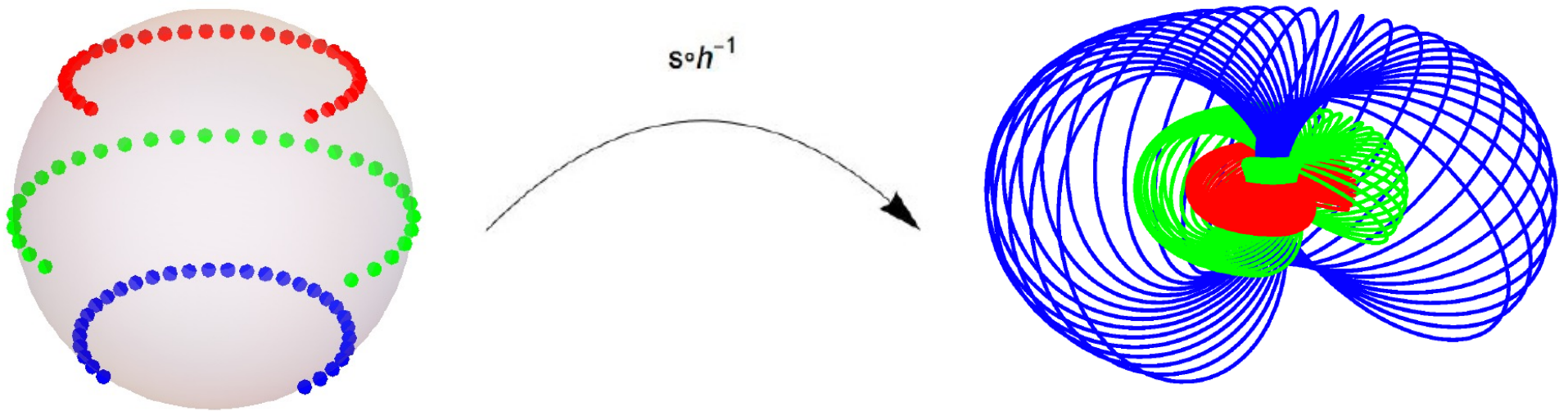
PREIMAGES: Fibers of the Hopf Map & Latitudes



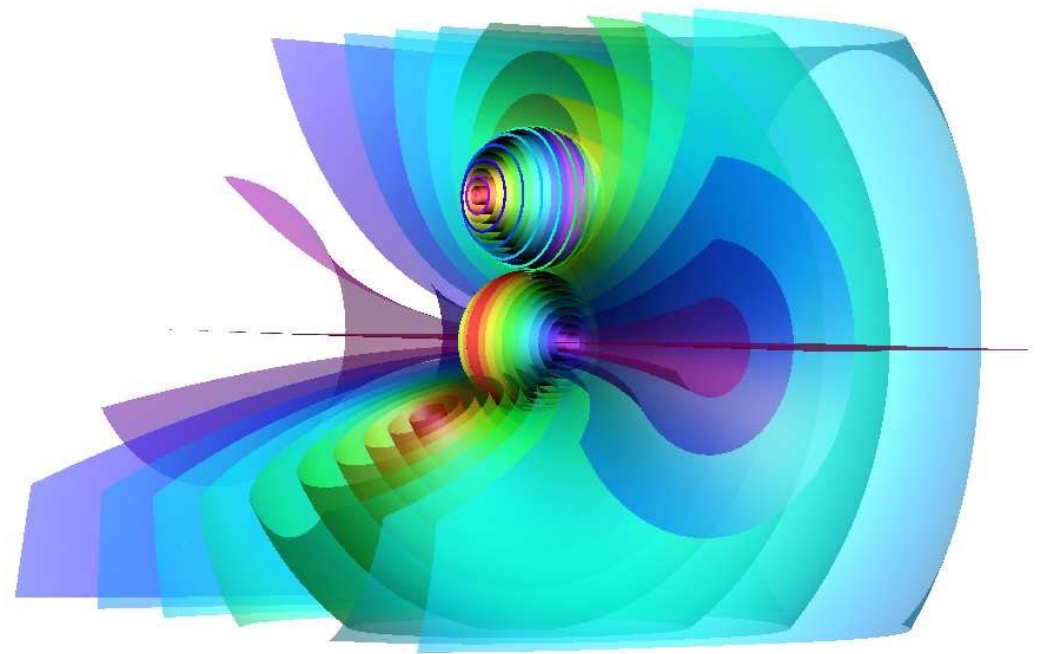
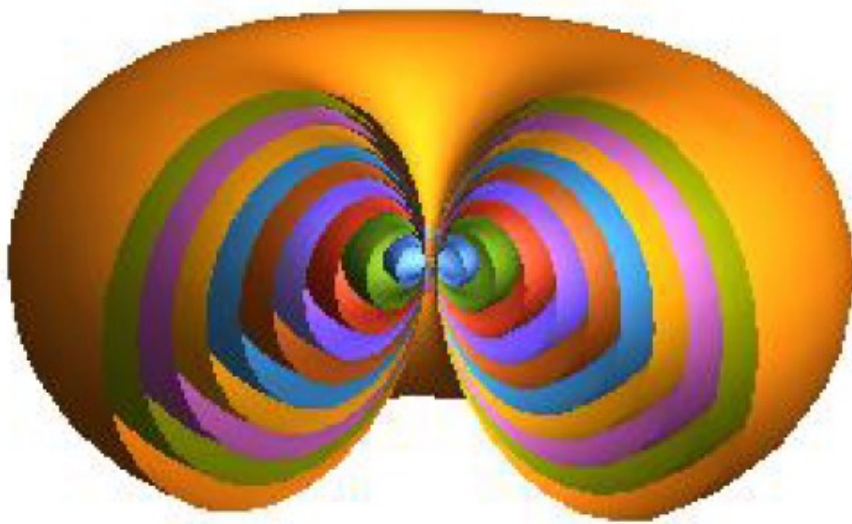
Preimage of a Latitude is a Dense Torus



The Hopf Fibration

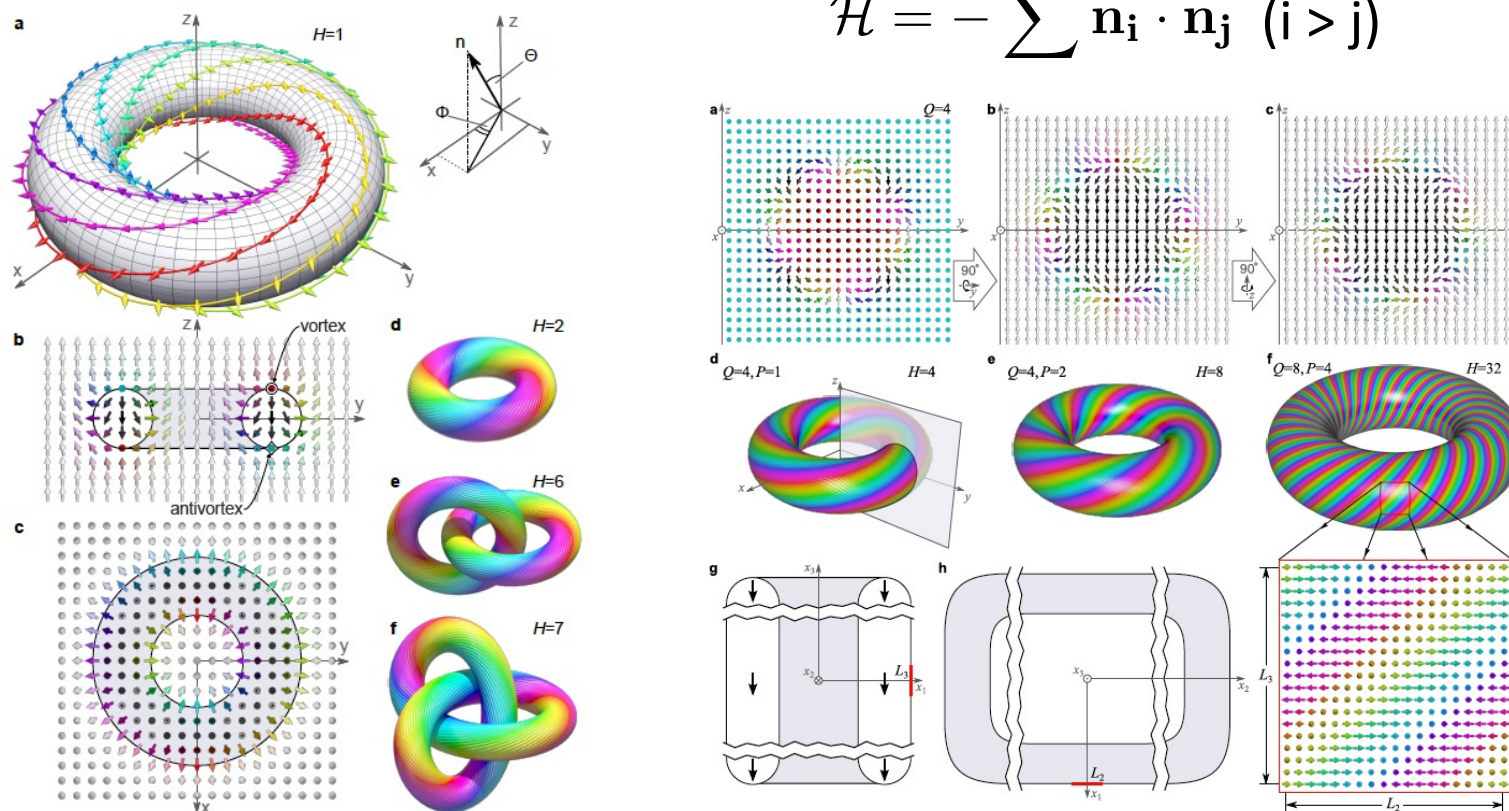


Nested Tori



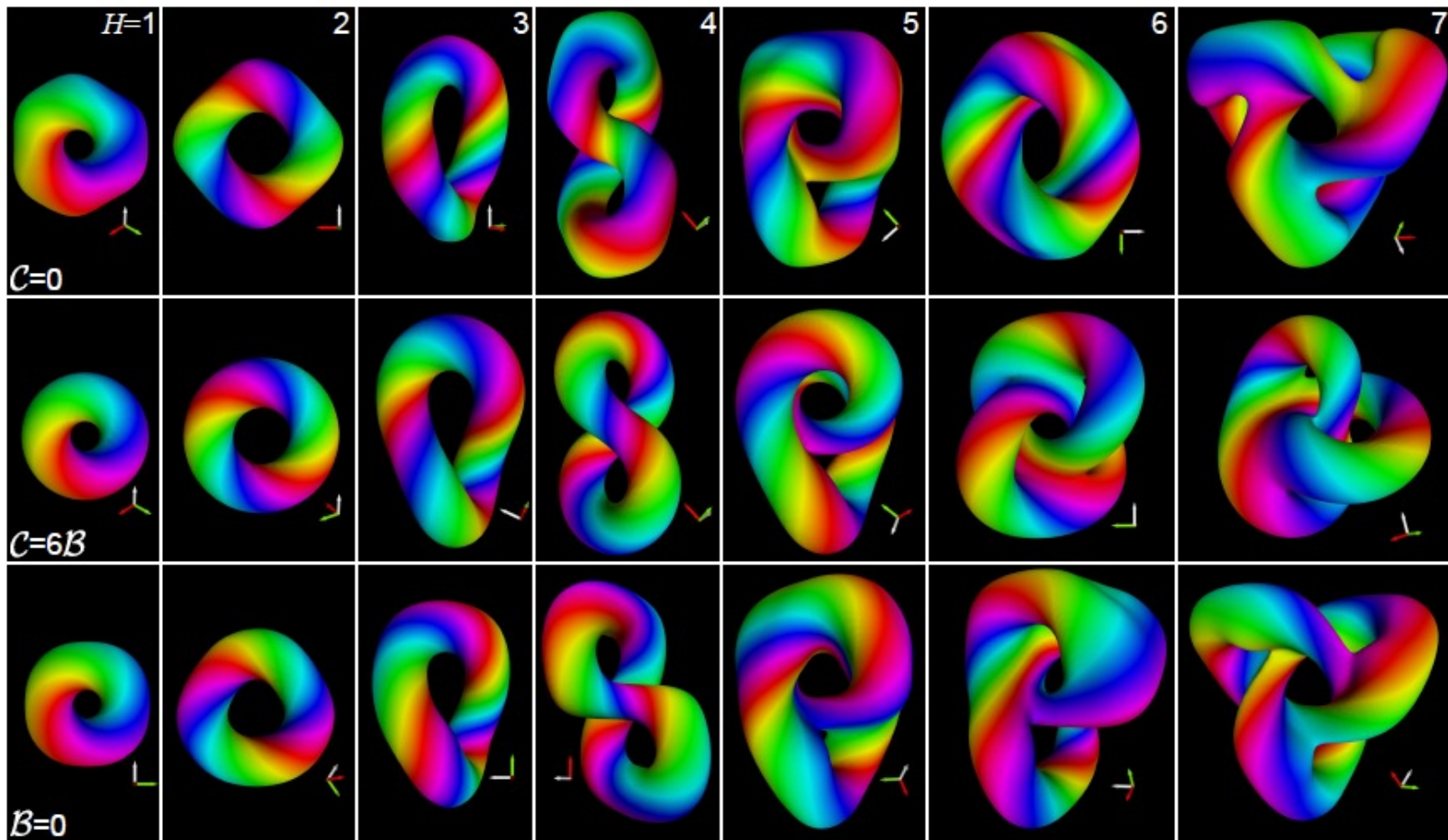
MAGNETIC HOPFIONS: Classical Heisenberg Model

$$\mathcal{H} = - \sum \mathbf{n}_i \cdot \mathbf{n}_j \quad (i > j)$$



F.N. Rybakov et al., arXiv:1904.00250 (AIP Adv. 2022)

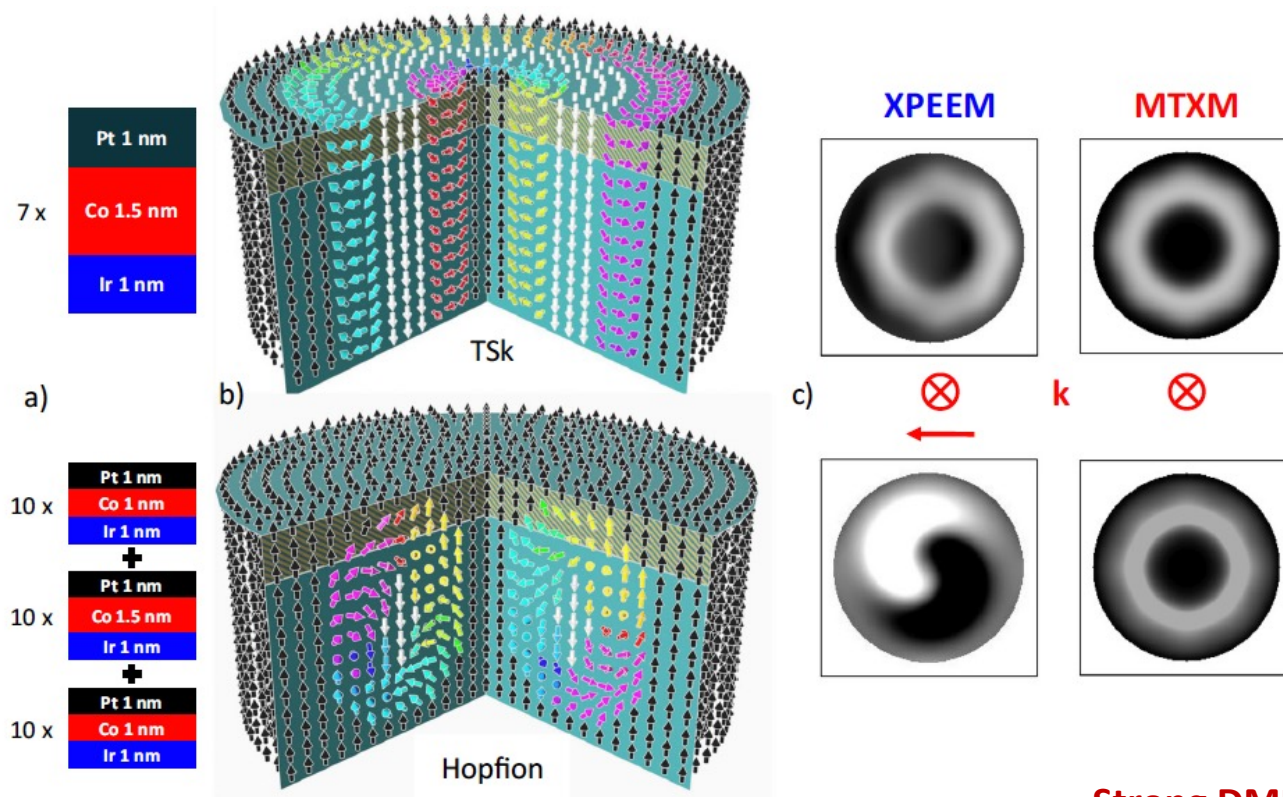
MORPHOLOGY OF MAGNETIC HOPFIONS: Hopf index H



M-field

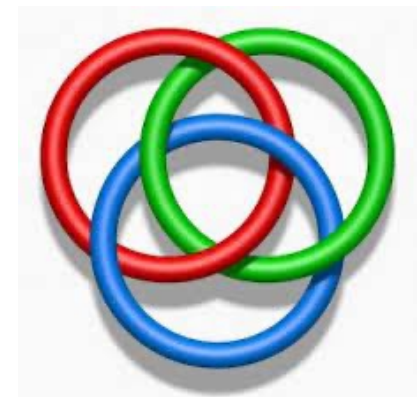
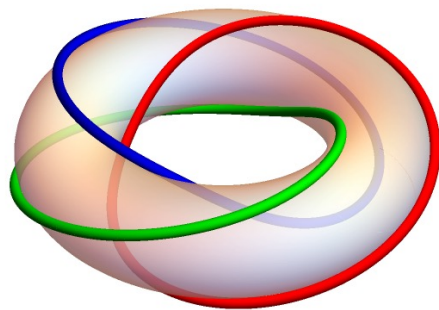
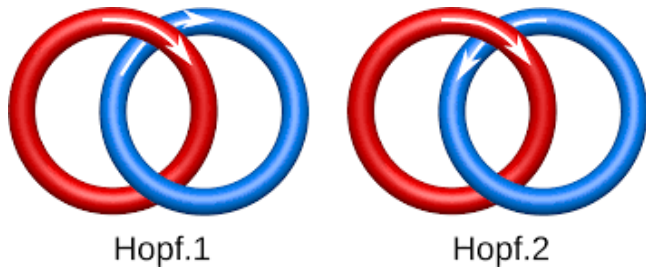
F. N. Rybakov
et al, arXiv:
1904.00250
(AIP Adv. 2022)

Experimental Observation of Hopfions: Magnetic Multilayers (Ir/Co/Pt)



N. Kent et al., Nature Commun. 12:1562 (2021)

Heinz Hopf (1894–1971)



Borromean rings

John H. C. Whitehead

$$H = -\frac{1}{8\pi^2} \iiint (\mathbf{A} \cdot \mathbf{B}) d^3\mathbf{r}$$
$$\mathbf{B} = \nabla \times \mathbf{A}$$



British mathematician:
One of the founders of
Homotopy Theory

Nov. 11, 1904 (**b. Chennai, India**) – May 8, 1960 (**d. Princeton, USA**)

Hopf Invariant (an integer):

$$H = -\frac{1}{4\pi} \int \int \int \vec{A} \cdot (\nabla \times \vec{A}) \, dx dy dz$$

$$\vec{B} = \nabla \times \vec{A}$$

J.H.C. Whitehead, Proc. Nat. Acad. Sci. (USA) **33**, 117 (1947)

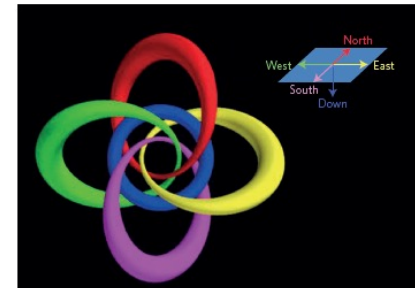
F. Wilczek and A. Zee, Phys. Rev. Lett. **51**, 2250 (1983)

H. Hopf, Mathematische Annalen **104**, 637 (1931): **Hopf map**

Special case of **Seifert map**: Herbert Seifert, Acta Mathematica **60**, 147 (1933)



HEINZ HOPF
(1894 – 1971)



Each colored curve is linked exactly once with any curve of a different color (**Paul Sutcliffe, Nature Mat. 2017**)

Hopfion Exact Solution: Anisotropic Heisenberg Ferromagnet

Hamiltonian:
$$E = \frac{J}{a} \int \int \int \left\{ \left(\frac{\partial \mathbf{S}}{\partial x} \right)^2 + \left(\frac{\partial \mathbf{S}}{\partial y} \right)^2 + \tilde{J}_3 \left(\frac{\partial \mathbf{S}}{\partial z} \right)^2 \right\} dx dy dz$$

Spin: $\mathbf{S} = [\sin \Theta \cos \Phi, \sin \Theta \sin \Phi, \cos \Theta]$

Cylindrical coordinates: (ρ, ϕ, z)

Consider solutions of the form: $\Theta = \Theta(\rho)$, $\Phi = (\alpha_0 \phi + \beta_0 z) + \Phi_0$

Hamiltonian:
$$E = \frac{J}{a} \int \int \int \left[\rho \left(\frac{\partial \Theta}{\partial \rho} \right)^2 + \frac{\alpha_0^2}{\rho} \sin^2 \Theta + \tilde{J}_3 \beta_0^2 \rho \sin^2 \Theta \right] d\rho d\phi dz$$

Euler-Lagrange equation:
$$\left[\rho \frac{\partial^2 \Theta}{\partial \rho^2} + \frac{\partial \Theta}{\partial \rho} \right] = \sin \Theta \cos \Theta \left[\frac{\alpha_0^2}{\rho} + \tilde{J}_3 \beta_0^2 \rho \right]$$

Hopfion Exact Solution: Anisotropic Heisenberg Ferromagnet

Assume inhomogeneous anisotropy: $\tilde{J}_3 = K_3 l^2 / \rho^2$ and $\rho = \rho_0 e^{\tilde{\rho}}$

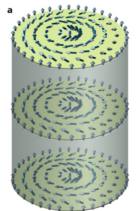
We obtain:
$$\frac{\partial^2(2\Theta)}{\partial \tilde{\rho}^2} = (\alpha_0^2 + K_3 l^2 \beta_0^2) \sin(2\Theta)$$

Impose periodic boundary conditions: $\alpha_0 = n$, $\beta_0 = 2\pi m/L$, $\mu^2 = \left(n^2 + K_3 \frac{4\pi^2 l^2}{L^2} m^2\right)$

Hopfion solution:

$$\begin{aligned} \theta(\rho) &= 2 \tan^{-1} \rho^{\pm\mu} \\ \Phi(\phi, z) &= n\phi + \frac{2\pi m}{L} z \end{aligned}$$

Twisted 2D skyrmion



$$\mathbf{S}(\rho, \phi, z) = [S_1, S_2, S_3] = \left[\frac{2\rho^{\pm\mu}}{1+\rho^{\pm 2\mu}} \cos(n\phi + 2\pi m z/L), \frac{2\rho^{\pm\mu}}{1+\rho^{\pm 2\mu}} \sin(n\phi + 2\pi m z/L), \frac{1-\rho^{\pm 2\mu}}{1+\rho^{\pm 2\mu}} \right]$$

Hopf invariant: $\boxed{\mathcal{H} = mn}$ = (winding no. of 2D skyrmion)(periodicity of closed loop in the 3rd dimension)

Energy of the Hopfion:

Solution: $\Theta = 2 \tan^{-1} (\rho/\rho_0)^\mu$; $\sin \Theta = \frac{2(\rho/\rho_0)^\mu}{[1 + (\rho/\rho_0)^{2\mu}]}$

Energy:
$$E = 2\pi \frac{JL}{a} \int_0^\infty \left[\rho \left(\frac{\partial \Theta}{\partial \rho} \right)^2 + \mu^2 \frac{\sin^2 \Theta}{\rho} \right] d\rho = \frac{8\pi JL}{a} |\mu|$$

Energy bound (over the **Hopf invariant H**): **Sublinear**

$$E \geq \left(\frac{8\pi JL}{a} 2^{1/4} \right) \sqrt{K_3} \frac{2\pi l}{L} \sqrt{nm} = c \sqrt{H} \quad c = 2^4 \pi^{3/2} K_3^{1/4} J \sqrt{lL}/a$$

For a skyrmion: Linear

Hopfion lattice solution

$$\cos \Theta = -k \operatorname{sn}(\mu \ln(\rho/\rho_0), k), \quad \sin \Theta = \operatorname{dn}(\mu \ln(\rho/\rho_0), k)$$

Periodicity: $\rho_p = \rho_0 \exp(2pK/\mu)$

Spin components:

$$\mathbf{S}(\rho, \phi, z) = \left[\operatorname{dn}(\mu \ln \frac{\rho}{\rho_0}, k) \cos(n\phi + 2\pi mz/L + \Phi_0), \operatorname{dn}(\mu \ln \frac{\rho}{\rho_0}, k) \sin(n\phi + 2\pi mz/L + \Phi_0), -k \operatorname{sn}(\mu \ln \frac{\rho}{\rho_0}, k) \right]$$

Energy:

$$E_{HL} = 2\pi \frac{JL\mu}{a} \int_0^K [2\operatorname{dn}^2(x, k) - k'^2] dx = 8\pi \frac{JL\mu}{a} \left[E(k) - \frac{k'^2}{2} K(k) \right]$$

Single Hopfion limit: $k \rightarrow 1$ (Elliptic modulus)

Calculation of the Hopf Invariant

Whitehead formula:
$$H = -\frac{1}{8\pi^2} \int \int \int \mathbf{A} \cdot \mathbf{B} \, dx \, dy \, dz$$

Emergent magnetic field:
$$B_1(y, z) = -\mathbf{S} \cdot (\mathbf{S}_y \times \mathbf{S}_z) = -\sin \Theta [\Theta_y \Phi_z - \Theta_z \Phi_y]$$

Cylindrical coordinates:

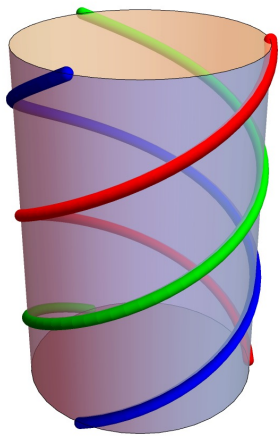
$$B_1 = \beta_0 [\cos \Theta]_\rho \sin \phi, \quad B_2 = -\beta_0 [\cos \Theta]_\rho \cos \phi, \quad B_3 = \frac{\alpha_0}{\rho} [\cos \Theta]_\rho$$

Vector field components:

$$A_1 = -\frac{\alpha_0}{\rho} [\pm 1 + \cos \Theta] \sin \phi, \quad A_2 = +\frac{\alpha_0}{\rho} [\pm 1 + \cos \Theta] \cos \phi, \quad A_3 = +\beta_0 \cos \Theta$$

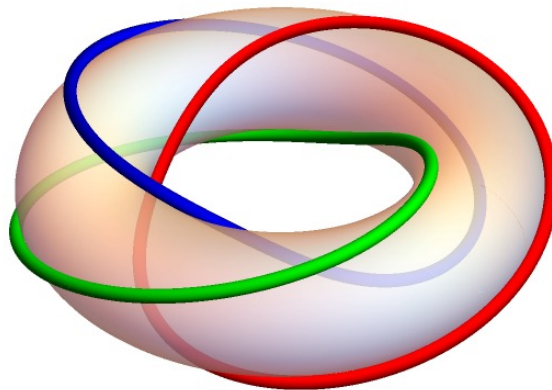
Hopf invariant:
$$H = \left(\mp \frac{1}{8\pi^2} \right) \alpha_0 \beta_0 \int_0^L \int_0^{2\pi} \int_0^\pi \sin \Theta \, d\Theta \, d\phi \, dz = mn$$

PREIMAGES

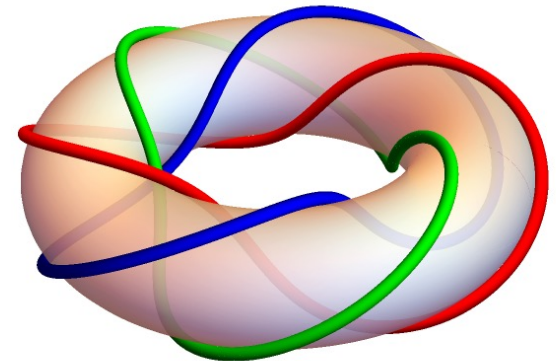


Cylinder

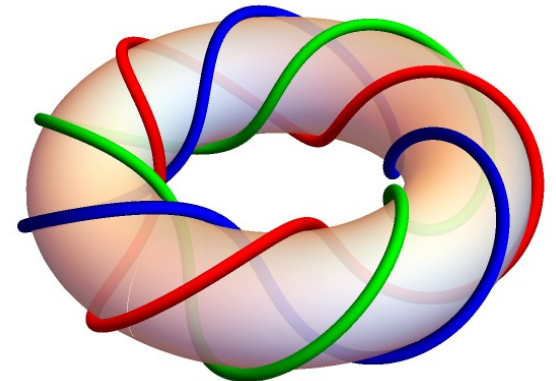
$m=1, n=1$ ($H=1$)



$m=1, n=2$ ($H=2$)

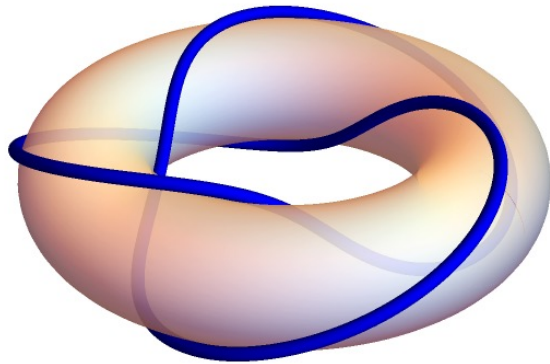


$m=1, n=3$ ($H=3$)

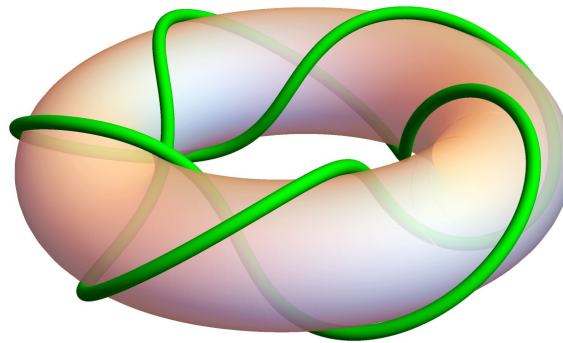


PREIMAGES

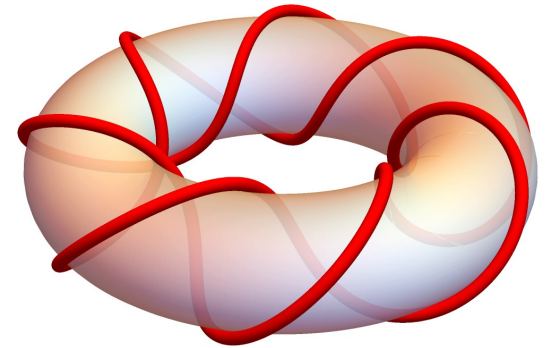
$m=2, n=3$ ($H=6$)



$m=1, n=5$ ($H=5$)

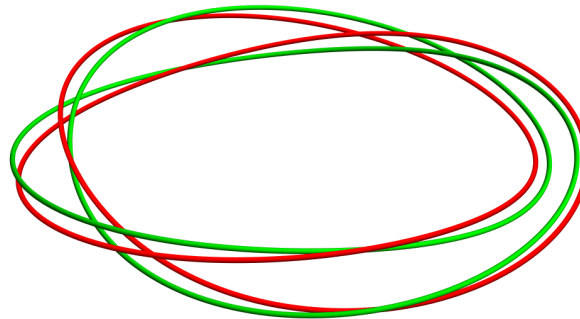
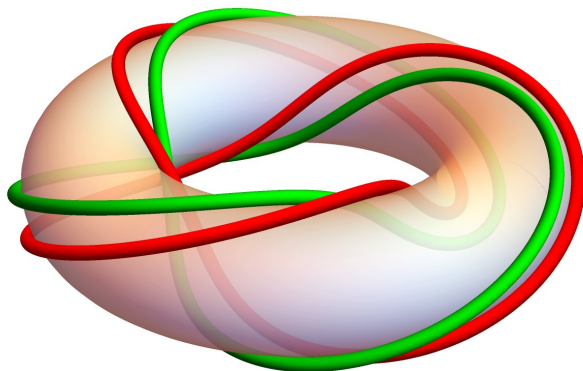


$m=1, n=7$ ($H=7$)



Trefoil

$m=2, n=3$ ($H=6$)

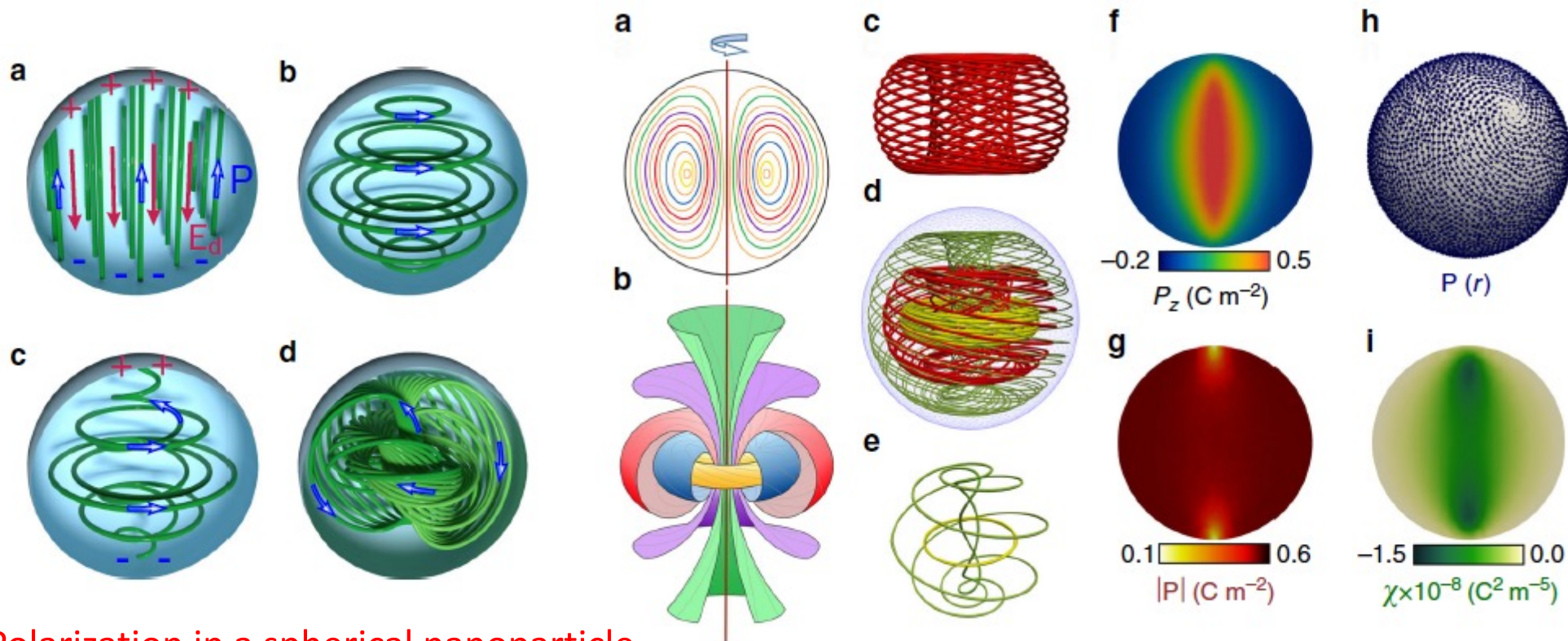


Densely covered torus



$m=n=1$

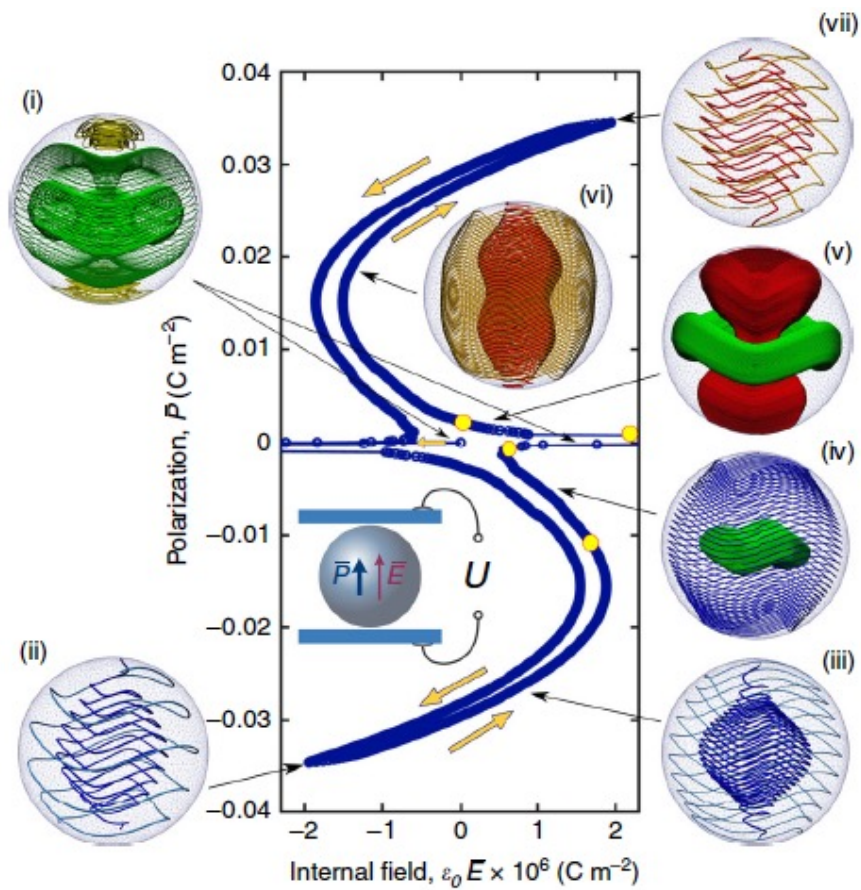
POLAR HOPFIONS (ferroelectrics)



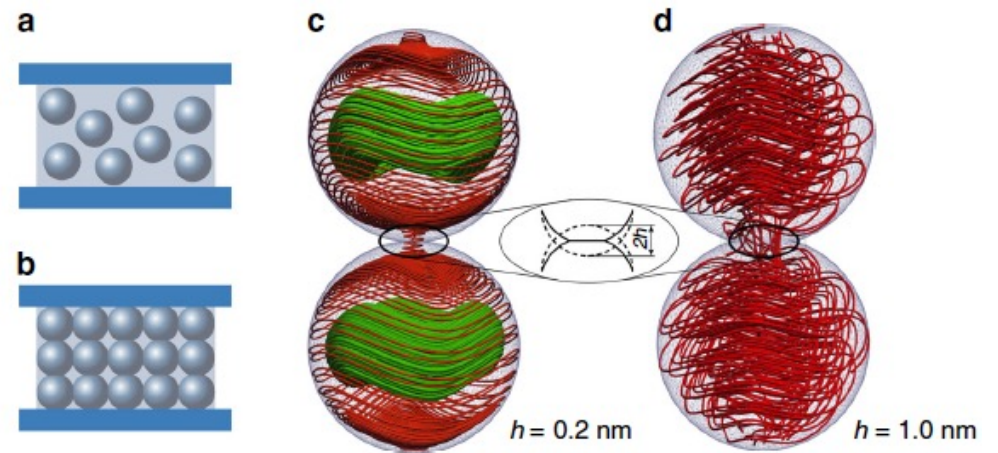
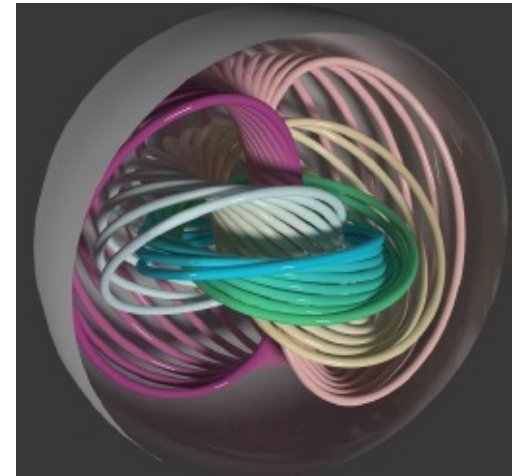
Polarization in a spherical nanoparticle

I. Luk'yanchuk et al., Nature Commun. 11, 2433 (2020)

POLAR HOPFIONS:



Field-induced topological states (PZT)



I. Luk'yanchuk et al., Nature Commun. 11, 2433 (2020)

Ginzburg-Landau Free Energy: Hopfions in Ferroelectrics

$$\begin{aligned} F = & \left[a_i(T) P_i^2 + a_{ij} P_i^2 P_j^2 + a_{ijk} P_i^2 P_j^2 P_k^2 \right]_{i \leq j \leq k} \\ & + \frac{1}{2} G_{ijkl} (\partial_i P_j) (\partial_k P_l) + (\partial_i \phi) P_i - \frac{1}{2} \varepsilon_0 \varepsilon_b [(\nabla \phi)^2 + \lambda^{-2} \phi^2] \\ & - C_{ijkl} Q_{klmn} u_{ij} P_m P_n + \frac{1}{2} C_{ijkl} u_{ij} u_{kl} \end{aligned}$$

Polarization induced electrostatic potential:

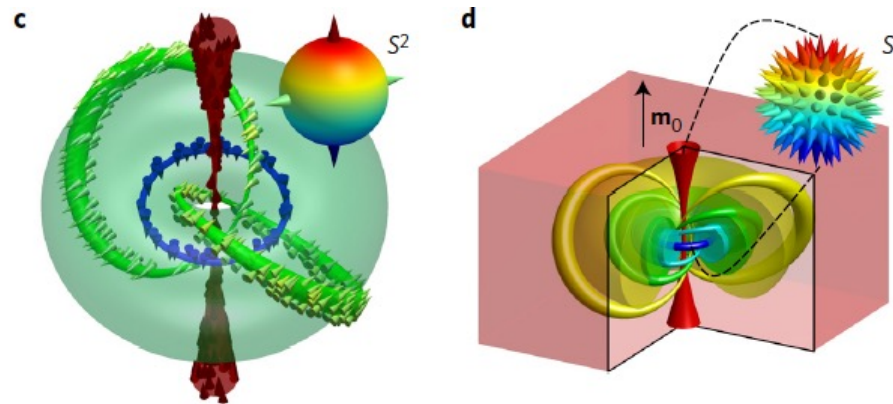
$$\varepsilon_0 \varepsilon_b [\nabla^2 - \lambda^{-2}] \phi = \partial_i P_i$$

Elastic strains:

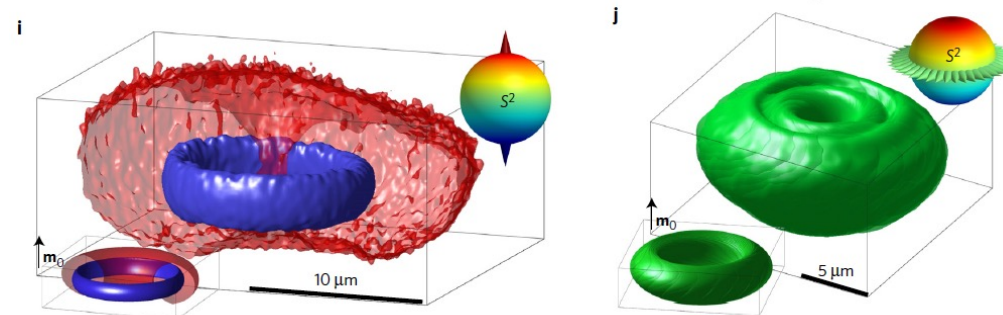
$$C_{ijkl} \partial_j (u_{kl} - Q_{klmn} P_m P_n) = 0$$

HOPFIONS in Liquid Crystals

Pre-images:
Linked circles

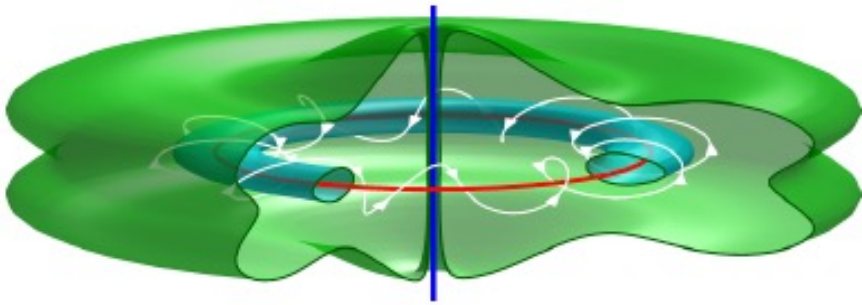


Pre-images of
N and S poles



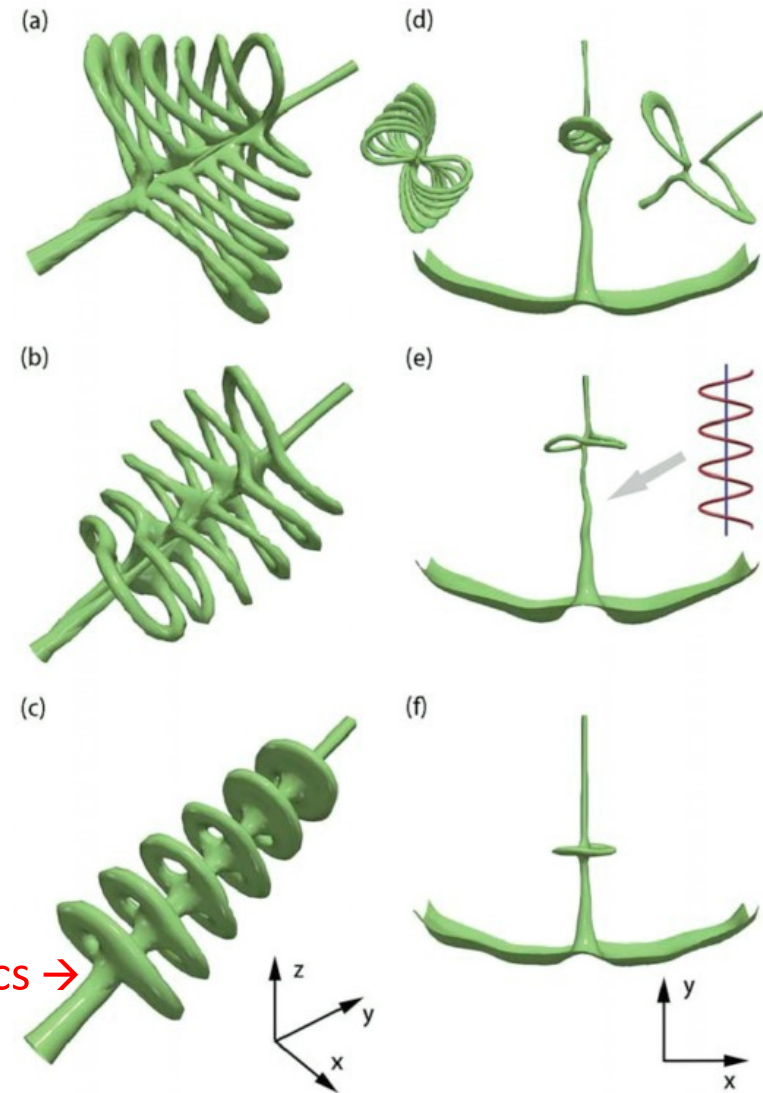
Ackermann and Smalyukh, Nature Mater. 16, 426 (2017)

HOPFIONS in rotating BEC:

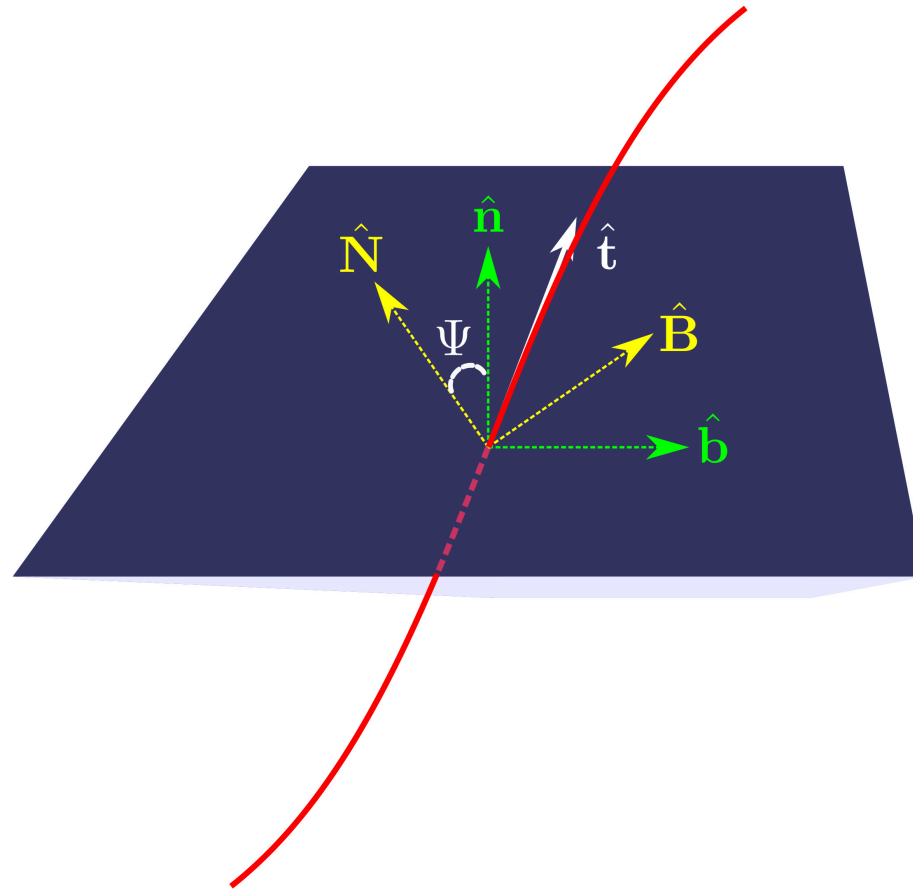


Y.M. Bidasyuk et al., Phys. Rev. A 92, 053603 (2015)

S. Zou et al., Phys. Fluids 33, 027105 (2021): Dynamics →



Space Curve Characterized by the Triad: normal (n), binormal (b), tangent (t)



arXiv:2304.06240

Frenet-Serret triad: Intrinsic twist

$$\int_M K dA = 2\pi\chi = 2\pi(2 - 2g)$$

Unit vector field: $\mathbf{t}(\mathbf{r}) = (\sin \Theta(\mathbf{r}) \cos \Phi(\mathbf{r}), \sin \Theta(\mathbf{r}) \sin \Phi(\mathbf{r}), \cos \Theta(\mathbf{r}))$

Rotated triad: $\mathbf{N} = \mathbf{n} \cos \Psi + \mathbf{b} \sin \Psi; \quad \mathbf{B} = \mathbf{t} \times \mathbf{N} = -\mathbf{n} \sin \Psi + \mathbf{b} \cos \Psi$

Frenet-Serret equations (rotated unit triad):

$$\mathbf{t}_x = (k_1 \cos \Psi) \mathbf{N} - (k_1 \sin \Psi) \mathbf{B}, \quad \mathbf{N}_x = -(k_1 \cos \Psi) \mathbf{t} + \tau_{T,1} \mathbf{B}, \quad \mathbf{B}_x = (k_1 \sin \Psi) \mathbf{t} - \tau_{T,1} \mathbf{N}$$

Curvature: $k_1(x) = |\mathbf{t}_x|, \quad \tau_1(x) = \mathbf{t} \cdot (\mathbf{t}_x \times \mathbf{t}_{xx}) / k^2$ Torsion

Berry connection and Berry phase (1D)

Total twist density:

$$\tau_{T,1}(x) = \tau_1(x) + \frac{d\Psi}{dx}$$

Intrinsic twist density

$$\tau_{T,1}(x) = \cos \Theta(x) \frac{d\Phi}{dx} + \frac{d\Psi}{dx}$$

Gauge dependent

Twist
Number:

$$\mathcal{N} = (1/2\pi) \int [\partial\psi/\partial x] dx \quad \mathbf{G}_x = \boldsymbol{\xi} \times \mathbf{G}$$

Darboux
Vector:

$$\boldsymbol{\xi} = \tau_{T,1} \mathbf{t} + k \sin \Psi \mathbf{N} + k \cos \Psi \mathbf{B}$$

Angular
velocity of
rotation

Berry
Phase:

$$\phi_B = \int \tau_{T,1}(x) dx = - \iint \sin \Theta d\Theta d\Phi + 2\pi \mathcal{N} = \Omega(C) + 2\pi \mathcal{N}$$

Gauge invariant

Solid angle

2D Manifold

Berry curvature:

$$\Omega_1(y, z) = \left[\frac{d\tau_{T,3}}{dy} - \frac{d\tau_{T,2}}{dz} \right]$$

Anholonomy density

$$\Omega_1(y, z) = \left[\frac{d\tau_3}{dy} - \frac{d\tau_2}{dz} \right] = \sin \Theta [\Theta_y \Phi_z - \Theta_z \Phi_y] \quad \text{Gauge invariant}$$

Total
anholonomy:

$$\begin{aligned} Q &= \frac{1}{4\pi} \int \int \Omega_1(y, z) dy dz = \frac{1}{4\pi} \int \int \mathbf{t} \cdot (\partial_y \mathbf{t} \times \partial_z \mathbf{t}) dy dz \\ &= -\frac{1}{4\pi} \int \int \sin \Theta [\Theta_y \Phi_z - \Theta_z \Phi_y] dy dz \quad \text{Berry phase} \end{aligned}$$

Winding number (Pontryagin index): S^2 to S^2 (integrand is torsion or nonplanarity)

3D Manifold

Berry Connection: $\mathbf{V} = (V_1, V_2, V_3) = (\tau_{T,1}, \tau_{T,2}, \tau_{T,3}) = \boldsymbol{\tau}_T$

Berry curvature: $\boxed{\boldsymbol{\Omega} = \nabla \times \boldsymbol{\tau}_T = \nabla \times \mathbf{V}}$ Divergence free

Volume integral: $h = \alpha \int \int \int \mathbf{V} \cdot \boldsymbol{\Omega} \, dx \, dy \, dz$

$$\boxed{h = \alpha \int \int \int \boldsymbol{\tau}_T \cdot (\nabla \times \boldsymbol{\tau}_T) \, dx \, dy \, dz}$$

$$h = \alpha \int \int \int \sin \Theta \left\{ \Psi_x [\Theta_y \Phi_z - \Theta_z \Phi_y] + \Psi_y [\Theta_z \Phi_x - \Theta_x \Phi_z] + \Psi_z [\Theta_x \Phi_y - \Theta_y \Phi_x] \right\} dx \, dy \, dz$$

Winding Number (S^3 to S^3)

Winding number:

$$h = \alpha \int \int \int \sin \Theta \, d\Theta \, d\Phi \, d\Psi$$

Torsion plus intrinsic twist

Normalized spinor representation: $\mathbf{t} = Z^\dagger \boldsymbol{\sigma} Z$

Hopf coordinate representation:

$$x_1 = \cos \Theta/2 \cos \xi_1; \quad x_2 = \cos \Theta/2 \sin \xi_1; \quad x_3 = \sin \Theta/2 \cos \xi_2; \quad x_4 = \sin \Theta/2 \sin \xi_2$$

$$z_1 = \cos \Theta/2 \exp i\xi_1; \quad z_2 = \sin \Theta/2 \exp i\xi_2$$

$$\Phi = (\xi_2 - \xi_1); \quad \Psi = -(\xi_1 + \xi_2)$$

Intrinsic twist

$$\xi_1 = -(\Psi + \Phi)/2; \quad \xi_2 = -(\Psi - \Phi)/2$$

$$Z^T = \exp(-i\Psi/2) \left(\cos \Theta/2 \exp(-i\Phi/2), \sin \Theta/2 \exp(i\Phi/2) \right)$$

Linking Number (S^3 to S^2) or Hopf Invariant

Whitehead formula: $H = -(1/8\pi^2) \iiint (\mathbf{A} \cdot \mathbf{B}) dx dy dz$ Linking number

Emergent magnetic field: $B_x = -\mathbf{t} \cdot (\partial_y \mathbf{t} \times \partial_z \mathbf{t})$

Emergent vector potential:

$$A_1 = [\pm 1 + \cos \Theta] \frac{\partial \Phi}{\partial x}; \quad A_2 = [\pm 1 + \cos \Theta] \frac{\partial \Phi}{\partial y}; \quad A_3 = \cos \Theta \frac{\partial \Phi}{\partial z}$$

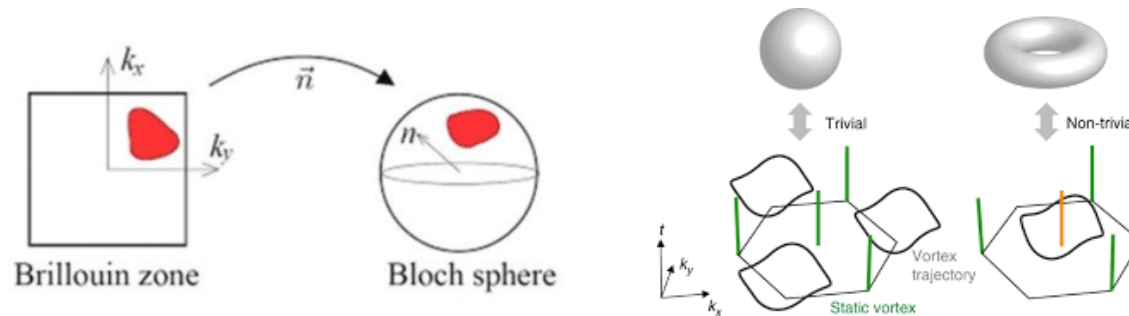
Gauge fixed (intrinsic twist): $\frac{\partial \Psi}{\partial x} = \pm \frac{\partial \Phi}{\partial x}; \quad \frac{\partial \Psi}{\partial y} = \pm \frac{\partial \Phi}{\partial y}; \quad \frac{\partial \Psi}{\partial z} = \frac{\partial \Phi}{\partial z}$

Mappings, Topological Invariants and Geometry

Dim.	Map	Topological Invariant	Geometric Quantity	Gauge	<u>Topo. soliton</u>
2D	$S^2 \rightarrow S^2$	W_2 (winding no.)	Torsion	None	Baby skyrmion
3D	$S^3 \rightarrow S^3$	W_3 (winding no.)	Torsion plus general twist	Arbitrary	Skyrmion
3D	$S^3 \rightarrow S^2$	H (linking no.)	Torsion plus fixed twist	Fixed	Hopfion
3D	$S^2 \times T^1 \rightarrow S^2$	H (linking no.)	Torsion plus fixed twist	Fixed	<u>Hopfion vortex</u>

Chern Insulators and Hopf Insulators

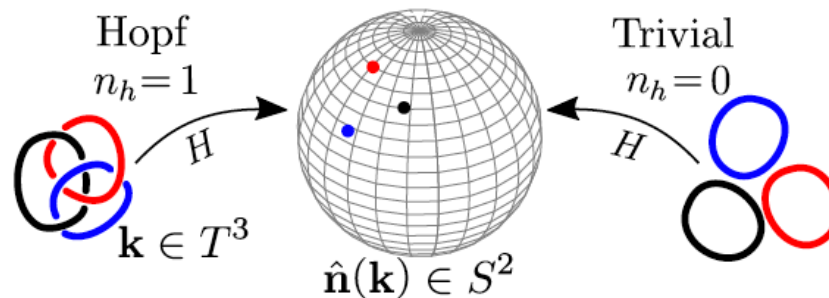
Map: $T^2 \rightarrow S^2$ (2D Brillouin Zone) **Chern number**



WSe₂ monolayer

M. Tarnowski et al.
Nature Commun.
10, 1038 (2019)

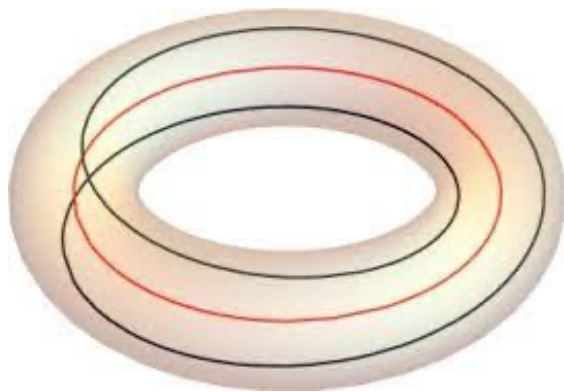
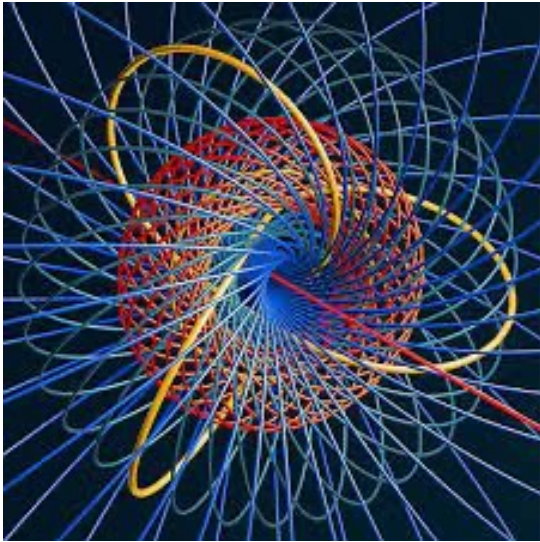
Map: $T^3 \rightarrow S^2$ (3D Brillouin Zone) **Hopf index**



Ultracold polar molecule ⁴⁰K⁸⁷Rb
(dipolar spin system)

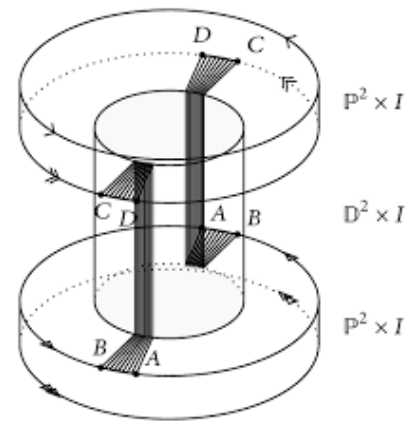
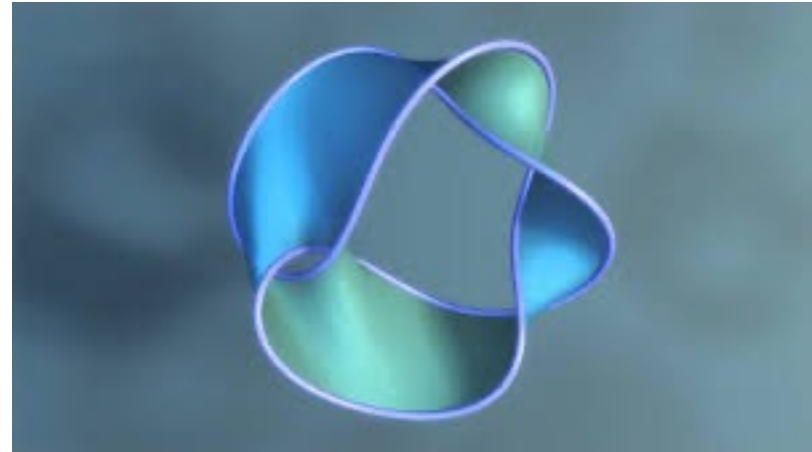
T. Schuster et al. PRL
127, 015301 (2021)

Seifert Fibration



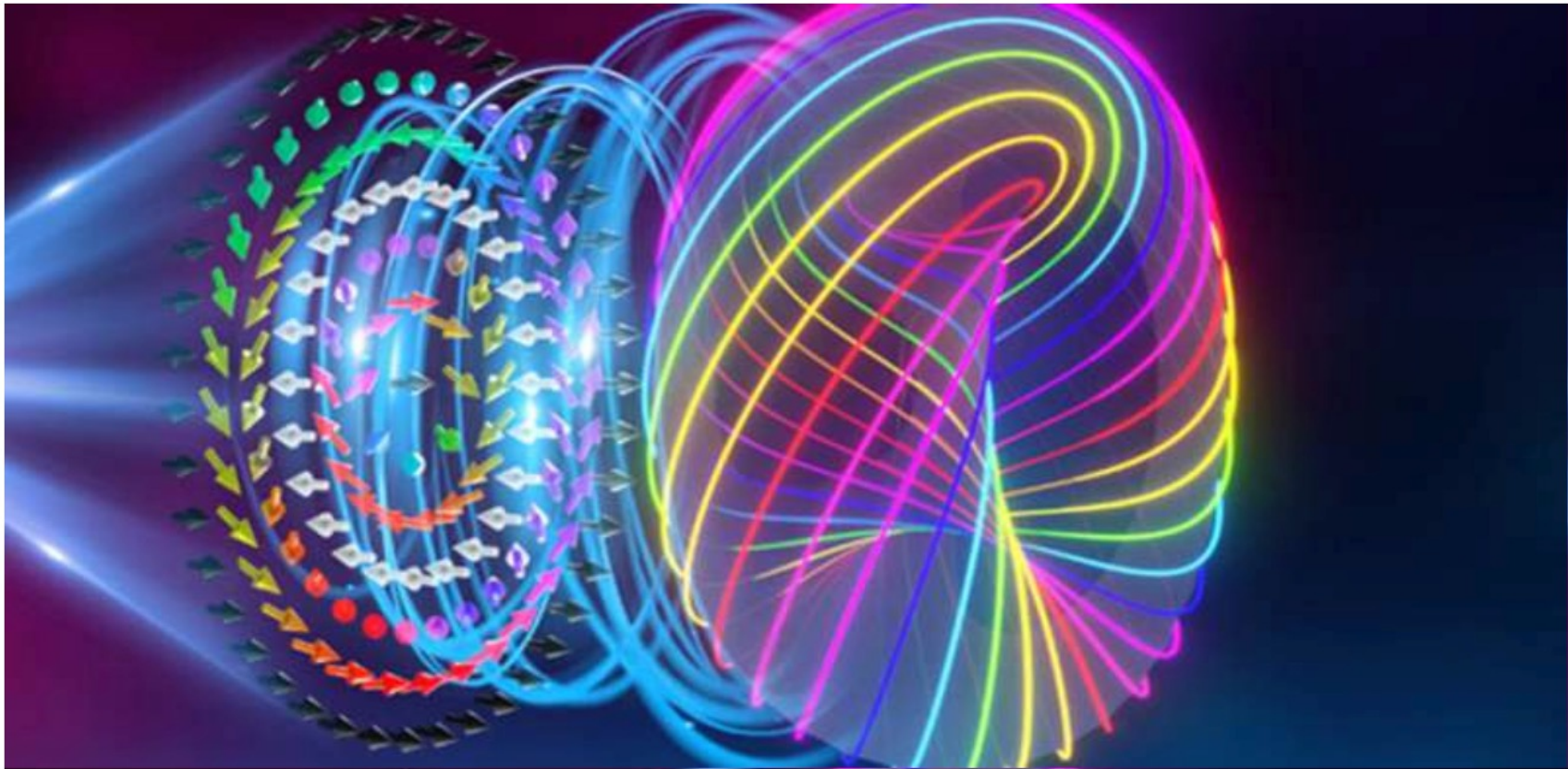
Seifert fibering
Operators in 3D

SEIFERT SURFACE

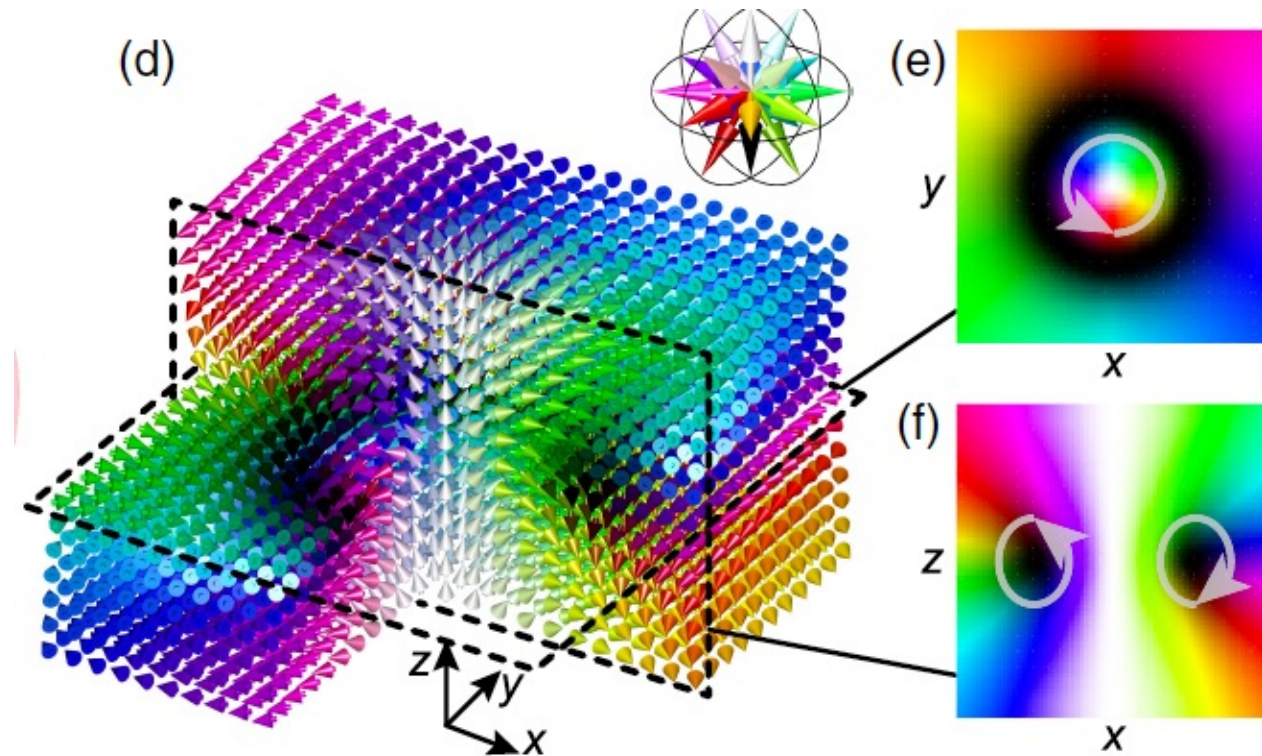


Seifert Fibered
Annulus on a
Nonorientable
3-Manifold

Photonic Hopfions: Structured Light



3D Spin Distribution in an Optical Hopfion

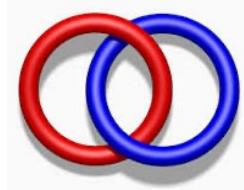


Y. Shen et al., Adv. Photonics **5**, 015001 (2023)

Homotopy Groups of the Spheres: S^n

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}
S^0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^1	$\mathbb{Z}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^2	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^3	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^4	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z} \times \mathbb{Z}_{12}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{24} \times \mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{120} \times \mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^5$
S^5	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{30}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{72} \times \mathbb{Z}_2$
S^6	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_{60}$	$\mathbb{Z}_{24} \times \mathbb{Z}_2$	$\mathbb{Z}_2^3$
S^7	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_{120}$	$\mathbb{Z}_2^3$
S^8	0	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z} \times \mathbb{Z}_{120}$

CONCLUSIONS:



- Growing **ubiquity** of (skyrmions and) hopfions in **condensed matter**.
- **Exact Hopfion** and Hopfion lattice **solutions** for anisotropic magnets.
- **Hopf invariant** $H = mn$, preimages are closed space curves (knots).
- Two preimage curves **link mn times**: **twisted** m times along z -axis.
- Anholonomy: **Intrinsic twist** necessary for a **topological invariant**.
- Hopfions in an **axial** (inhomogeneous) **magnetic field**; **Photonics**.
- Properties and **application of hopfions** a **new materials frontier!**